Fire Radiative Power Dynamics in Nepal Himalayan Forests: Spatiotemporal Drivers and Machine Learning Characterization of Fire Combustion Intensity (2012–2024)

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Research Article

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Abstract

Forest and vegetation fires in Nepal Himalayan landscape pose significant threats to biodiversity, watershed function, and forest dependent communities across the Hindu Kush Himalaya (HKH), yet fire combustion intensity as distinct from fire occurrence has never been systematically characterised or modelled for this region. This study presents the first machine learning characterisation of Fire Radiative Power (FRP) dynamics across Nepal four ecological zones (Terai, Siwaliks, Middle Hills, High Mountains) using a 13-year VIIRS active fire archive spanning January 2012 to December 2024. Spatiotemporal clustering of 569,136 quality controlled VIIRS S-NPP and NOAA-20 fire detections using a grid based aggregation approach (0.25° cells, 2 day gap tolerance) yielded 11,595 discrete fire events characterised by FRP statistics, spatial extent, duration, and ecological zone. Fire activity was strongly concentrated in the pre monsoon season, with April alone accounting for 55.6% of all events. The High Mountains zone contributed 27.2% of all fire events a substantially higher proportion than previously documented revealing an intense high elevation fire regime in trans Himalayan districts that has been systematically underrepresented in coarser resolution satellite records. Marked inter annual anomalies were identified: 2015 recorded the fewest events of the study period (n = 442, approximately 50% below the 13 year mean), coinciding with the Gorkha earthquake (Mw 7.8), while 2021 recorded the highest event count (n = 1,198), consistent with documented extreme fire activity across the HKH.

Environmental drivers of fire intensity were quantified using Random Forest regression with five fold spatial block cross validation across 23 predictor variables 18 standard meteorological, terrain, fuel, and temporal predictors, and five novel terrain wind interaction features engineered from ERA5-Land and SRTM inputs. The model achieved a mean cross validated *R^2 of 0.161 (± 0.083) and RMSE of 4.10 (± 0.92) MW*, with within zone performance ranging from *$R^2 = 0.683$*in the High Mountains to *$R^2 = 0.757$* in the Siwaliks. Air temperature (permutation importance = 0.064), cropland fraction (0.062), elevation (0.054), and vapour pressure deficit (0.051) ranked as the four most important predictors globally. Critically, the engineered drought fuel interaction index combining vapour pressure deficit with broadleaf forest fraction ranked fifth (0.044), exceeding raw wind speed (rank 13; 0.019), demonstrating that the interaction between atmospheric drying and fuel type modulates fire intensity more strongly than instantaneous wind speed across Nepal forest dominated Middle Hills landscape. XGBoost modelling confirmed the Random Forest driver hierarchy ($R^2 = 0.151$) and SHAP attribution analysis revealed a systematic elevation gradient in feature importance: elevation SHAP contributions increased monotonically from the Terai (0.062) to the High Mountains (0.093), establishing elevation mediated climate as the primary structural organiser of fire intensity variation across Nepal ecological gradient. The marginal predictive contribution of terrain wind interaction features (+0.001 R^2) is interpreted as a resolution constraint arising from ERA5-Land 11 km atmospheric grid rather than evidence against the physical importance of terrain wind coupling in Nepal complex terrain a limitation that higher resolution mesoscale atmospheric inputs could address in future work. Physical consistency of FRP based intensity classifications was confirmed through Sentinel-2 post fire burn severity analysis of 111 fire events (2017–2024), with positive dNBR recorded across all FRP intensity classes; the observed negative FRP dNBR correlation (Spearman $\rho = -0.239$, $p = 0.012$) is

attributed to a vegetation density confound arising from the elevation stratification of Nepal fire landscape rather than measurement inconsistency.

These findings provide the first quantitative driver attribution framework for fire combustion intensity in the HKH, with direct implications for fire danger forecasting, community forest management, and climate change vulnerability assessment across Nepal and the broader Hindu Kush Himalaya region.

1. Introduction

1.1 Fire in the Hindu Kush Himalaya: Global Significance and Regional Knowledge Gaps

Forest and vegetation fires in the Hindu Kush Himalaya (HKH) represent one of the most consequential and least studied fire regimes on Earth. Spanning eight countries across an area of approximately 3.4 million km² and encompassing the world highest mountain range, the HKH sustains a disproportionate share of global biodiversity, freshwater resources, and carbon stocks while supporting the livelihoods of approximately 240 million people in the mountain zone and nearly two billion people downstream who depend on HKH water towers for agricultural, domestic, and industrial water supply (Immerzeel et al., 2020; Wester et al., 2019). Recurrent fire disturbance in this landscape degrades forest cover, accelerates soil erosion, reduces snowmelt runoff regulation, impairs watershed function, and releases substantial quantities of greenhouse gases and aerosols into the atmosphere consequences that reverberate well beyond the burned areas themselves (Vadrevu et al., 2019; Giglio et al., 2013; Miehe et al., 2021). Despite this broad significance, the HKH fire regime remains poorly characterised relative to comparably important fire landscapes in Australia, the western United States, the Amazon, and the boreal zone, where decades of satellite based fire research have produced well validated fire behaviour models, driver attribution frameworks, and operational fire danger forecasting systems (Chuvieco et al., 2022; Flannigan et al., 2016).

The fundamental challenge in characterising fire behaviour in the HKH is the exceptional complexity of the landscape itself. The region encompasses the most pronounced elevation gradient on Earth rising from subtropical lowlands at approximately 60 m above sea level to the summit of Mount Everest at 8,849 m within a horizontal distance of less than 200 km creating a compressed mosaic of climatic regimes, fuel types, and terrain configurations that interact in ways that no single fire behaviour model developed for flatter terrain can adequately represent (Körner, 2003; Sharma et al., 2022). Steep slope gradients exceeding 30° to 40° are commonplace in the Siwaliks and Middle Hills zones, generating strong terrain wind coupling through slope valley circulation systems that can dramatically accelerate fire spread and intensity in ways that are poorly captured by standard meteorological inputs (Sharples, 2009; Sharples et al., 2012). The South Asian Monsoon system, which delivers approximately 80% of annual precipitation between June and September, creates a sharply defined fire season confined to the pre-monsoon months of March through May a ten week window of low humidity, high temperature, strong winds, and depleted fuel moisture that concentrates fire activity into a period of exceptional intensity and ecological impact (Paudel et al., 2024; Devkota et al., 2023).

1.2 Nepal Fire Regime: What is Known and What Remain Unknown

Nepal exemplifies the broader HKH fire challenge at a national scale. With a land area of approximately 147,181 km² spanning four ecologically distinct zones the subtropical Terai lowlands, the Siwaliks foothills, the Middle Hills, and the High Mountains Nepal experiences a diverse array of fire types ranging from intentional agricultural burning in the lowland plains to wind driven forest fires in the steep hill terrain and episodic extreme fires in subalpine juniper and shrubland communities of the trans Himalayan districts (Bhujel et al., 2022; Pokhrel et al., 2026). Forest fires in Nepal cause significant economic losses through timber destruction, non timber forest product loss, and damage to community forestry systems that govern approximately 1.8 million hectares of the country forest area an institutional framework that represents one of Asia most successful forest governance experiments and whose resilience under recurrent fire disturbance has been inadequately studied (Devkota et al., 2023; Paudel et al., 2024).

Existing fire research for Nepal has advanced considerably over the past decade but has been largely constrained to fire occurrence mapping and spatial risk assessment frameworks. Bhujel et al. (2022) examined the temporal and spatial distribution of fire detections from MODIS and VIIRS data across Nepal, identifying agricultural burning and intentional ignition as the primary drivers of fire occurrence in the Terai and Siwaliks. Devkota et al. (2023) applied deep learning to MODIS derived fire occurrence data to map fire pattern and vulnerability at the national scale, demonstrating the potential of machine learning approaches for Nepal fire research but focusing on fire probability rather than fire intensity. Paudel et al. (2024) developed a MaxEnt based fire risk assessment for Gorkha District, integrating terrain, climate, and land cover data to identify high risk zones at the district scale. While these studies represent important advances in understanding where fires occur in Nepal, none has characterised the combustion intensity of fire events that is, not just whether a fire occurred, but how intensely it burned, what environmental conditions drove that intensity, and how intensity varies systematically across Nepal ecological gradient.

This gap between fire occurrence characterisation and fire intensity characterisation is scientifically significant because occurrence and intensity are driven by partially distinct sets of environmental controls and have partially distinct ecological and policy consequences. Fire occurrence is primarily controlled by ignition probability which in Nepal is overwhelmingly anthropogenic and therefore linked to agricultural calendar, land use practice, and access while fire intensity, once ignition occurs, is primarily controlled by fuel moisture state, wind conditions, terrain configuration, and fuel type (Pausas & Keeley, 2021; Chuvieco et al., 2022). High intensity fires produce disproportionate ecological damage relative to their extent, generating stand replacing tree mortality, severe soil heating, accelerated erosion, and long term changes in forest composition and regeneration dynamics that low intensity fires do not (Coop et al., 2020; Keeley, 2009). From a fire management perspective, the distinction between extensive fire seasons and intensive fire seasons between many fires and very intense fires demands fundamentally different response strategies and resource allocations (Jolly et al., 2015; Abatzoglou et al., 2019).

1.3 Fire Radiative Power as a Fire Behaviour Metric

Fire Radiative Power (FRP), measured in megawatts and retrieved from the thermal infrared channels of satellite sensors, provides a physically grounded proxy for fire combustion intensity that addresses the limitations of occurrence based fire characterisation. FRP is retrieved through a radiometric inversion of the radiant energy flux from active fire pixels and is proportional to the instantaneous rate of fuel consumption a direct link to fire line intensity that makes it a physically meaningful measure of combustion behaviour rather than simply a detection indicator (Wooster et al., 2005; Kaufman et al., 1998). Through integration over the fire duration, FRP yields fire radiative energy (FRE), which is proportional to total biomass consumed and total smoke emissions connecting event level fire intensity measurements to landscape scale carbon flux and air quality consequences (Wooster et al., 2005; Kaiser et al., 2012). Critically, FRP responds to the same environmental drivers as fire rate of spread and fireline intensity wind speed, fuel moisture, slope gradient, and fuel type making it a behaviour metric that can be analysed within existing fire behaviour theoretical frameworks while remaining entirely satellite derived and therefore applicable without any ground based measurement infrastructure (Rothermel, 1972; Schroeder et al., 2014).

The Visible Infrared Imaging Radiometer Suite (VIIRS) 375 m active fire product (VNP14IMG), derived from the Suomi National Polar orbiting Partnership satellite and the complementary NOAA-20 satellite, represents the current state of the art in satellite fire detection and FRP retrieval, offering substantially improved detection capability for small fires and better spatial resolution than its MODIS predecessor (Schroeder et al., 2014). The combination of VIIRS S-NPP (operational from 2012) and VIIRS NOAA-20 (operational from 2018) provides approximately four satellite overpasses per day over Nepal from 2018 onward a temporal density sufficient to characterise diurnal fire cycles and capture peak intensity observations during the afternoon fire window representing a previously unexploited observational resource for HKH fire research (Schroeder & Giglio, 2016).

Despite its demonstrated utility as a fire behaviour proxy in fire research across North America, Europe, Australia, and Africa (Andela et al., 2019; de Groot et al., 2015; Giglio et al., 2013), FRP has not been systematically applied to characterise fire combustion intensity in the HKH. No published study has used FRP as a primary response variable for driver attribution or machine learning modelling in Nepal or any adjacent HKH country. This represents a fundamental gap in regional fire science that the present study addresses directly.

1.4 Machine Learning for Fire Behaviour Driver Attribution

The complexity of interactions among meteorological, terrain, fuel, and temporal drivers of fire intensity in heterogeneous landscapes makes machine learning particularly well suited for fire behaviour driver attribution capable of capturing non linear interactions and high dimensional predictor relationships that conventional statistical models cannot represent (Coffield et al., 2019; Jain et al., 2020). Random Forest regression and gradient boosting algorithms have been applied with considerable success to fire behaviour problems in other regions, including the prediction of fire size in Canada (de Groot et al., 2015), fire spread rate in Australia (Cruz et al., 2015), and fire severity in the western United States (Parks et al., 2019; Coop et al., 2020). However, existing machine learning fire behaviour studies have been developed overwhelmingly for topographically simple or moderately complex terrain, where standard meteorological predictors wind speed, temperature, relative humidity, drought index adequately capture the dominant fire driving conditions without terrain specific correction.

In complex mountain terrain such as Nepal Siwaliks and Middle Hills, the effectiveness of standard meteorological predictors is limited by the mismatch between the spatial scale at which atmospheric models resolve wind fields and the scale at which terrain wind coupling operates at the fire event level. Slope wind interactions, valley channelling, and aspect driven solar radiation effects on fuel moisture all operate at spatial scales of 1–5 km below the resolution of global reanalysis products requiring explicit terrain wind feature engineering to capture in a machine learning framework (Sharples, 2009; Rotach et al., 2017). No study has developed or tested such terrain wind interaction features for fire behaviour prediction in the HKH, and their contribution to fire intensity prediction relative to standard predictors remains unquantified for this region.

The recent development of SHapley Additive exPlanations (SHAP) as a framework for model agnostic interpretability has substantially increased the scientific value of machine learning approaches for driver attribution by providing rigorous, theoretically grounded decompositions of model predictions into individual predictor contributions at the observation level (Lundberg & Lee, 2017). SHAP has been applied to fire behaviour problems in several recent studies, enabling the identification of key driver thresholds, interaction effects, and spatial heterogeneity in driver importance across landscape gradients (Jain et al., 2020; Coffield et al., 2019). The application of SHAP to zone stratified fire intensity attribution in the HKH context revealing how driver hierarchies shift across Nepal elevation gradient represents both a methodological contribution and a substantive scientific advance for the region.

1.5 Study Objectives

Against this background of fire regime significance, knowledge gaps, and methodological opportunity, the present study pursues four interconnected objectives:

Objective 1. To construct the first comprehensive, quality controlled fire event database for Nepal (2012–2024) from VIIRS S-NPP and NOAA-20 active fire detections, characterising each event by fire radiative power statistics, spatial extent, temporal duration, dominant ecological zone, and FRP based intensity classification.

Objective 2. To quantify the relative importance of meteorological, terrain, fuel type, and temporal predictors of fire combustion intensity (FRP) across Nepal four ecological zones using Random Forest regression with five fold spatial block cross validation and permutation based feature importance.

Objective 3. To engineer and evaluate five novel terrain wind interaction features upslope wind component, aspect solar azimuth alignment, valley confinement index, wind slope interaction, and drought fuel interaction index and test their contribution to fire intensity prediction using XGBoost with SHAP attribution analysis, assessing whether terrain mediated processes improve prediction beyond standard meteorological and terrain predictors.

Objective 4. To assess the physical plausibility of FRP based fire intensity classifications through a Sentinel-2 post fire burn severity consistency check, and to examine the correspondence between the Canadian Fire Weather Index as implemented by ICIMOD for the HKH region and observed fire combustion intensity across Nepal ecological zones.

Together these objectives deliver a quantitative characterisation of what drives dangerous fires in Nepal Himalayan landscape that is directly applicable to fire danger forecasting, forest management planning, and climate change vulnerability assessment in the HKH region.

2. Study Area

Nepal (26°N-30.5°N, 80°E-88.2°E) is a landlocked country in South Asia spanning approximately 147,181 km², situated along the southern slopes of the central Himalayan range between India to the south and the Tibet Autonomous Region of China to the north (Fig. 1). Despite its relatively small size, Nepal encompasses one of the most pronounced elevation gradients on Earth, rising from the subtropical Terai plains at approximately 60 m above sea level in the south to the summit of Mount Everest at 8,849 m above sea level in the north. This extreme topographic relief produces a corresponding diversity of climatic regimes, ecological zones, and forest types, making Nepal a uniquely complex environment for fire behaviour research.

For the purposes of this study, Nepal landscape is stratified into four ecologically and topographically distinct zones following the classification system established by the Government of Nepal, Department of Forests and Soil Conservation also known as DFRS. The Terai zone occupies the southernmost lowland plains below approximately 300 m above sea level and is characterized by subtropical riverine and sal (*Shorea robusta*) forests, tall grasslands, and seasonally flooded wetlands. The Terai is Nepal most densely populated and agriculturally productive region, and fire in this zone is predominantly anthropogenic in origin, driven by land clearing, post harvest burning, and intentional ignition for grassland management (Bhujel et al., 2022). The Siwaliks (or Chure Hills) zone, extending from approximately 300m to 1,500 m above sea level, comprises the outermost foothills of the Himalayan arc and supports tropical and subtropical mixed forests dominated by *Shorea robusta*, *Terminalia*, and *Dalbergia* species. The steep, south facing slopes of the Siwaliks experience pronounced dry season moisture stress, creating conditions favorable for fire ignition and spread during the pre monsoon season (Paudel et al., 2024).

The Middle Hills zone, spanning roughly 1,500m to 3,500m above sea level, represents Nepal most ecologically diverse terrain and contains the majority of the country's human settlements. Forest cover in this zone includes subtropical broadleaf forests of *Castanopsis* and *Schima* at lower elevations, transitioning to temperate oak (*Quercus spp*.) and rhododendron (*Rhododendron* spp.) forests at higher elevations. Human forest interfaces are frequent throughout the Middle Hills, and fire ignition is closely linked to agricultural burning cycles, particularly the clearing of slash and burn cultivation areas and the burning of dry undergrowth for pasture management. Our dataset places the largest share of fire events (*n = 3,696*) in this zone, consistent with the high human forest interface and the abundance of flammable fuel types. The High Mountains zone above 3,500 m above sea level includes subalpine and alpine ecosystems characterized by conifer forests of *Abies spectabilis* and *Betula utilis* at lower subalpine elevations, transitioning to juniper (*Juniperus spp*.) scrubland, alpine meadows, and permanent snow and ice at the highest elevations. Although fire frequency is generally lower in this zone, the presence of highly flammable juniper and alpine shrub communities in trans Himalayan districts such as Mustang, Dolpo, and Upper Karnali sustains a non negligible fire regime (Pokhrel et al., 2026).

Nepal climate is dominated by the South Asian Monsoon system, which delivers approximately 80% of the country annual precipitation between June and September (Bhuju et al., 2007, as cited in Devkota et al., 2023). The dry season from October through May, and particularly the pre monsoon months of March, April, and May, corresponds to the period of peak fire activity across all ecological zones. During this season, low relative humidity, elevated temperatures, strong pre monsoon westerly and northwesterly winds, and depleted soil moisture create conditions conducive to fire ignition and spread. In our 13-year detection dataset (2012–2024), April alone accounted for 55.6% of all VIIRS fire detections, with March and May contributing an additional 19.0% and 14.7% respectively, confirming the dominance of the pre monsoon fire window across Nepal ecological gradient.

The complex terrain of Nepal Siwaliks and Middle Hills has particular significance for fire behavior. Steep slope gradients (*frequently exceeding 30°*), narrow valley systems, and strong diurnal wind patterns driven by slope valley circulation create highly heterogeneous fire environments in which fire intensity and spread can vary dramatically over short distances. This terrain complexity is a central motivation for the present study: existing fire danger indices and satellite based fire characterization approaches developed for flatter terrain may not adequately capture fire combustion dynamics in Nepal topographically complex landscape, and the role of terrain wind interactions in modulating fire radiative power remains unquantified for the Hindu Kush Himalaya (HKH) region.

Fire events were distributed across all four ecological zones, with the highest spatial density observed in the Middle Hills and Siwaliks zones where agricultural forest interfaces are most extensive (Fig. 1). Extreme and High intensity events (red and dark orange symbols) are concentrated in the Middle Hills and High Mountains zones, consistent with the steep terrain, pre monsoon drought conditions, and flammable subalpine fuel types characteristic of these elevations.

3. Data Sources

3.1 VIIRS Active Fire Detections

The primary fire detection dataset used in this study is the Visible Infrared Imaging Radiometer Suite (VIIRS) 375 m active fire product (VNP14IMG), derived from the Suomi National Polar orbiting Partnership (S-NPP) satellite launched in October 2011, and the complementary NOAA 20 satellite product (VJ114IMG) available from April 2018 onward. The VIIRS 375 m active fire algorithm was developed by

Schroeder et al. (2014) and represents a substantial improvement over its MODIS predecessor, offering enhanced detection of smaller fires and improved spatial resolution critical for fragmented fire landscapes such as Nepal Middle Hills. The algorithm uses dual middle and thermal infrared channels centered at *3.75 μm and 11.45 μm* to identify fire affected pixels through contextual temperature anomaly detection, assigning each detection a confidence classification of high (h), nominal (n), or low (l) and a Fire Radiative Power (FRP) value in megawatts (MW).

Fire detection data were accessed through the NASA Fire Information for Resource Management System (FIRMS) archive download portal for the period January 2012 through December 2024. Two product streams were downloaded: VIIRS S-NPP Standard Processing (SP) Collection 2 covering the full study period, and VIIRS NOAA-20 SP Collection 2 covering April 2018 to December 2024. The dual satellite archive from 2018 onward provides approximately four overpasses per day over Nepal (two from S-NPP and two from NOAA-20), compared to approximately two overpasses per day in the single satellite period (2012–2017). This difference in temporal sampling density is explicitly accounted for in all trend analyses (Section 4).

A spatial filter was applied to retain detections within Nepal bounding box (80.0°E–88.2°E, 26.0°N–30.5°N). Quality filtering retained only nominal (n) and high (h) confidence detections with positive FRP values and fire type code 0 (vegetation fires), excluding industrial thermal anomalies, offshore detections, and volcanoes (type codes 2 and 3). Following quality filtering, the merged S-NPP and NOAA-20 dataset comprised 569,136 individual fire detections across the 13-year study period.

3.2 Meteorological Reanalysis Data

Meteorological variables driving fire combustion intensity were sourced from the ERA5-Land hourly reanalysis dataset produced by the European Centre for Medium Range Weather Forecasts (ECMWF) within the framework of the Copernicus Climate Change Service (Muñoz Sabater et al., 2021). ERA5 Land provides hourly estimates of land surface variables at 0.1° × 0.1° resolution (~ 11 km) globally from 1950 to present, produced by integrating the ECMWF land surface model with downscaled ERA5 atmospheric forcing that includes an elevation correction for near surface thermodynamic state variables. This elevation correction is particularly relevant for Nepal highly variable topography, where lapse rate corrected temperature and moisture fields better represent conditions at fire affected elevations than uncorrected reanalysis products.

The following ERA5 Land variables were extracted for each fire detection at the time and location of the detection: 10 metre U and V components of wind velocity (u10, v10), from which wind speed and direction were derived; 2 metre air temperature (t2m); 2 metre dewpoint temperature (d2m), used to calculate vapour pressure deficit (VPD); and volumetric soil water content in the 0 to 7 cm surface layer (swvl1) as a proxy for surface fuel moisture. VPD was calculated as the difference between saturated and actual vapour pressure derived from t2m and d2m following standard meteorological formulae. A 30 day antecedent precipitation deficit index was also computed from ERA5 Land total precipitation, capturing pre fire drought conditions that influence fuel dryness.

3.3 Digital Elevation Model and Terrain Derivatives

Topographic variables were derived from the Shuttle Radar Topography Mission (SRTM) Version 3 digital elevation model (DEM) at 1 arc second (~ 30 m) resolution (Farr et al., 2007). The SRTM was acquired during an 11 day Space Shuttle Endeavour mission in February 2000 using dual antenna synthetic aperture radar interferometry, producing the most complete near global high resolution topographic dataset available at the time of this study. The SRTM Version 3 (SRTM Plus) product used here incorporates void filling from ASTER GDEM2 and GMTED2010 sources and was accessed through the USGS Earth Explorer portal and processed via Google Earth Engine (GEE).

Four terrain variables were computed from the SRTM DEM using GEE's built in terrain analysis functions: (1) elevation (m above sea level); (2) slope gradient (degrees), derived using the maximum downslope gradient algorithm; (3) aspect (degrees from north, 0 to 360°), representing the compass direction of the slope face; and (4) Topographic Position Index (TPI), computed as the difference between a pixels elevation and the mean elevation within a 1 km radius, providing a continuous measure of topographic context ranging from valleys (strongly negative TPI) to ridgelines and peaks (strongly positive TPI). Terrain variables were resampled to 0.1° resolution to match the ERA5 Land grid prior to feature extraction.

3.4 Land Cover and Fuel Type

Fuel type proxies were derived from the European Space Agency (ESA) WorldCover 10 m global land cover product. Both the 2020 (v100) and 2021 (v200) editions were used, providing the highest spatial resolution globally consistent land cover classification available for the study period (Zanaga et al., 2021; Zanaga et al., 2022). The World Cover product classifies land surface into 11 classes using a machine learning algorithm trained on Sentinel 1 SAR and Sentinel 2 multispectral data, with a global overall accuracy of approximately 75% for the 2020 product. For this study, World Cover classes were remapped to five fire relevant fuel categories: broadleaf forest, needleleaf forest, mixed/shrubland, grassland/herbaceous, and cropland/other, following fuel classification frameworks established for Himalayan fire risk studies (Devkota et al., 2023; Paudel et al., 2024).

3.5 Fire Weather Index

Fire danger context was provided by the Canadian Fire Weather Index (FWI) system outputs produced by the International Centre for Integrated Mountain Development (ICIMOD) through its Hindu Kush Himalaya Integrated Water and Hazard Assessment Tool (HIWAT) project. The FWI system is a meteorologically based index framework that computes fuel moisture codes for fine fuels, duff, and large woody debris, as well as fire behaviour indices for rate of spread, buildup, and overall fire danger (Van Wagner, 1987). ICIMOD's HIWAT FWI outputs are specifically calibrated for the HKH regional context and represent the most regionally appropriate fire danger product available for Nepal. In this study, daily FWI values were extracted at each fire event's location and start date to examine the correspondence between fire danger rating and observed fire combustion intensity.

Table 1
Data sources used in this study

Dataset	Source / Product	Spatial Res.	Temporal Coverage	Use in Study
VIIRS S-NPP 375m	NASA FIRMS VNP14IMG SP C2	375 m	Jan 2012 – Dec 2024	Primary fire detections + FRP
VIIRS NOAA-20 375m	NASA FIRMS VJ114IMG SP C2	375 m	Apr 2018 – Dec 2024	Secondary fire detections + FRP
ERA5-Land Hourly	ECMWF / Copernicus CDS	0.1° (~ 11 km)	2012–2024, hourly	Wind, temperature, VPD, soil moisture
SRTM DEM v3	NASA / USGS EarthExplorer	30 m (~ 1 arc-sec)	February 2000 (static)	Elevation, slope, aspect, TPI
ESA WorldCover	ESA / Copernicus (Zenodo)	10 m	2020 (v100), 2021 (v200)	Fuel type / land cover proxy
FWI (HIWAT)	ICIMOD HKH HIWAT	~ 10 km	2012–2024, daily	Fire danger context / validation

4. Methods

4.1 Fire Detection Pre-processing and Quality Filtering

Individual VIIRS active fire detections from both S-NPP and NOAA-20 platforms were subjected to a three stage quality filtering protocol prior to analysis. First, a spatial filter retained only detections within Nepal national boundary bounding box (80.0°–88.2°E, 26.0°–30.5°N). Second, a confidence filter retained nominal (confidence = 'n') and high (confidence = 'h') quality detections only, excluding low confidence detections (confidence = 'l') that carry elevated commission error rates. Third, a fire type filter retained only vegetation fire detections (type = 0), excluding industrial thermal anomalies (type = 2), offshore detections (type = 3), and other non vegetation sources. Detections with missing or zero FRP values were also excluded as these indicate failed radiometric retrievals. Following quality filtering, the merged dataset comprised 569,136 fire detections with a median FRP of 2.2 MW and maximum FRP of 626.1 MW across the 13 year study period.

Column name standardization was applied to harmonize variable names across the S-NPP and NOAA-20 product streams. Acquisition dates were parsed from FIRMS standard YYYY-MM-DD format to R Date objects, and acquisition times were converted from HHMM integer format to decimal hours to enable temporal analysis. Each detection was assigned to one of four ecological zones (Terai, Siwaliks, Middle Hills, High Mountains) based on its latitude using threshold values consistent with Nepal national ecological classification framework. All pre processing was conducted in R version 4.3 using the *tidyverse* and *lubridate* packages.

4.2 Spatiotemporal Fire Event Definition

Individual VIIRS active fire detections represent instantaneous satellite observations of active fire pixels and do not inherently correspond to discrete fire events. To define ecologically meaningful fire events for driver analysis and machine learning modelling, a grid based spatiotemporal aggregation approach was developed and applied to the full 13-year detection archive.

Nepal fire active land area was partitioned into a regular grid of 0.25° × 0.25° cells (approximately 28 km × 28 km at Nepal latitude range), yielding 253 active grid cells across the study domain. This grid resolution was selected to satisfy three competing requirements: spatial specificity sufficient to distinguish fires in adjacent valley systems or ridge complexes; compatibility with the ERA5 Land 0.1° meteorological extraction grid through nearest neighbour resampling; and approximate correspondence with Nepal district level fire management administrative boundaries (Paudel et al., 2024). Within each grid cell, fire detections were sorted chronologically and grouped into discrete fire events using a temporal gap tolerance of two days. A new fire event was initiated when the elapsed time since the most recent detection within the same grid cell exceeded two days, consistent with gap tolerance parameters applied in comparable spatiotemporal fire clustering studies in complex terrain (Schroeder et al., 2014; Andela et al., 2019). This tolerance accommodates missed satellite overpasses resulting from cloud obscuration or smoke interference while ensuring that genuinely independent ignition episodes separated by more than two days are treated as distinct events.

An initial pool of 26,057 grid cell event groups was subjected to a quality filter retaining only events satisfying at least one of the following criteria: two or more active detection days (*n_active_days* ≥ 2), confirming multi-day persistence; or five or more individual detections (*n_detections* ≥ *5*), confirming spatial extent sufficient to exclude single pixel radiometric anomalies. This filter reduced the event pool to 11,595 quality controlled fire events representing 96.6% of all quality filtered VIIRS detections, confirming that the removed marginal events account for a negligible fraction of total fire radiative energy. Each event was characterised by centroid coordinates, spatial extent, temporal duration, dominant ecological zone, dominant satellite source, percentage of nocturnal detections, and the full suite of FRP statistics (mean, median, maximum, total, and standard deviation). Fire events were classified into four intensity categories based on peak FRP: Low (FRP_max < 15 MW), Moderate (15–50 MW), High (50–150 MW), and Extreme (FRP_max ≥ 150 MW), following intensity thresholds established empirically from the distribution of the 13-year Nepal dataset and consistent with FRP based fire intensity frameworks applied in comparable mountainous fire environments (Wooster et al., 2005; de Groot et al., 2015).

The grid based aggregation approach was selected over density based spatial clustering (DBSCAN) after systematic parameter testing revealed that DBSCAN produced spatially indefensible results under Nepal high fire density conditions (Ester et al., 1996). During the peak pre monsoon fire season, simultaneous ignitions across all ecological zones generate densely connected detection networks in which DBSCAN chains geographically distant fires into single clusters spanning the majority of the study domain. The largest DBSCAN cluster identified during parameter testing (eps = 0.10°, minPts = 3) contained 6,822 detections spanning 7.9° of longitude and 3.3° of latitude on a single day during the record 2021 fire season a physically implausible single fire event. The grid based method resolves this by enforcing strict spatial boundaries at the grid cell level, ensuring fires in different catchments or ridge systems are always treated as independent events regardless of detection density in adjacent cells.

4.3 Environmental Feature Extraction:

For each of the 11,595 fire events, environmental predictor variables were extracted at the event centroid and start date using Google Earth Engine (Gorelick et al., 2017). Feature extraction was performed independently for three data sources and subsequently merged on event identifier.

Meteorological variables were extracted from the ERA5 Land hourly reanalysis dataset (Muñoz-Sabater et al., 2021) at a spatial resolution of 0.1° (~ 11 km). For each event, the following variables were extracted at two diurnal time points: 06:00 UTC (approximately 11:45 Nepal Standard Time, representing pre fire morning conditions) and 09:00 UTC (approximately 14:45 NST, representing peak fire activity conditions, consistent with the afternoon wind driven fire period documented for the HKH region; Paudel et al., 2024). Extracted variables included the zonal and meridional components of 10 metre wind velocity (u10, v10), from which wind speed and meteorological wind direction were derived; 2 metre air temperature (T2m); and volumetric soil water content in the 0–7 cm surface layer (swvl1) as a proxy for surface fuel moisture. Vapour pressure deficit (VPD, kPa) was computed from 2 metre air temperature and dewpoint temperature following standard thermodynamic formulae (Allen et al., 1998): *VPD = e_s(T) − e_a(T_d), where e_s and e_a* are the saturated and actual vapour pressures at temperature T and dewpoint T_d respectively, computed using the Magnus approximation.

Terrain variables were derived from the Shuttle Radar Topography Mission Version 3 digital elevation model at 1 arc second (~ 30 m) resolution (Farr et al., 2007), processed in Google Earth Engine. Four terrain variables were computed: elevation (m above sea level); slope gradient (degrees), using the maximum downslope gradient algorithm; aspect (degrees from north, 0–360°); and Topographic Position Index (TPI), defined as the difference between a pixel's elevation and the mean elevation within a 1 km radius kernel, providing a continuous index of topographic context ranging from valleys (strongly negative TPI) to ridgelines and peaks (strongly positive TPI; De Reu et al., 2013). Terrain variables were extracted at both the event centroid (90 m sampling scale) and the mean over the containing 0.25° grid cell to capture both local and landscape scale topographic context.

Fuel type variables were derived from the ESA WorldCover 10 m global land cover product, 2021 edition (Zanaga et al., 2022). WorldCover classes were remapped to five fire relevant fuel categories (broadleaf forest, shrubland, grassland, cropland, other) following fuel classification frameworks established for Himalayan fire studies (Devkota et al., 2023). The fractional cover of each fuel category within the containing 0.25° grid cell was computed as the primary fuel composition predictor, supplemented by the modal fuel class at the event centroid point.

4.4 Terrain Wind Interaction Feature Engineering

Standard meteorological and terrain predictors available in existing fire behaviour datasets were supplemented by five engineered features designed to capture terrain wind interactions specific to Nepal complex topography. These features are not available in existing fire danger products for the HKH region and represent the principal methodological contribution of the present study.

The upslope wind component was computed as the scalar projection of the afternoon wind vector onto the direction of steepest slope ascent at each fire event location:

$$W\_upslope = u10 \cdot \sin(\alpha) \cdot \sin(\beta) + v10 \cdot \cos(\alpha) \cdot \sin(\beta)$$

where α is slope aspect (radians) and β is slope gradient (radians). Positive values indicate wind directed upslope, which accelerates fire spread and intensity through increased convective coupling and pre heating of unburned fuels ahead of the fire front (Rothermel, 1972; Sharples, 2009).

The aspect solar azimuth alignment index quantified the angular difference between slope aspect and the solar azimuth angle at Nepal latitude (~ 28°N) during the pre monsoon fire peak (approximately 155° at 12:00 NST). Slopes facing toward the sun receive greater direct insolation, producing lower fuel moisture and higher fire radiative power under equivalent meteorological conditions (Polade et al., 2017).

The valley confinement index was derived from TPI as: $V = \max(0, -TPI) / (|TPI| + 1)$, producing a continuous 0 to 1 index in which values approaching 1 indicate deeply confined valley terrain. Valley confinement is associated with fire channelling, intensification through reduced atmospheric mixing, and the development of fire-induced local wind systems (Sharples et al., 2012).

The wind slope interaction term was computed as the product of afternoon wind speed and the sine of slope gradient: $W\_slope = |wind\_speed| \cdot \sin(\beta)$. This term explicitly represents the combined effect of wind speed and slope steepness on fire intensity that multiplicative interactions in terrain fire behaviour models predict (Rothermel, 1972; Andrews, 2018).

The drought fuel interaction index was computed as: $D\_fuel = VPD \times f\_broadleaf$, where $f\_broadleaf$ is the fractional broadleaf forest cover of the grid cell. This index captures the synergistic effect of atmospheric drying (VPD) on fuel moisture in the broadleaf forest fuel type that dominates Nepal Middle Hills fire landscape, following the fuel moisture FRP relationship established by Dennison et al. (2006) and Yebra et al. (2013).

Table 2
Environmental predictor variables used in machine learning models

#	Predictor variable	Description	Data source	Units	Type
W1	*era5_wind_speed_afternoon*	10 m wind speed at 09:00 UTC (~ 14:45 NST, peak fire hour)	ERA5-Land hourly	m s^{-1}	Standard
W2	*era5_wind_speed_morning*	10 m wind speed at 06:00 UTC (~ 11:45 NST)	ERA5-Land hourly	m s^{-1}	Standard
W3	*era5_wind_dir_afternoon*	Wind direction at 09:00 UTC, meteorological convention	ERA5-Land hourly	degrees	Standard
C1	*era5_temp_c*	2 m air temperature at 09:00 UTC	ERA5-Land hourly	°C	Standard
C2	*era5_vpd_kpa*	Vapour pressure deficit at 09:00 UTC [es(T) − ea(Td)]	ERA5-Land hourly	kPa	Standard
C3	*era5_soil_moisture*	Volumetric soil water content, 0–7 cm layer	ERA5-Land hourly	m^3 m^{-3}	Standard
T1	*srtm_elevation_m*	Elevation at event centroid	SRTM v3, 30 m	m a.s.l.	Standard
T2	*srtm_slope_deg*	Slope gradient at event centroid (max. downslope)	SRTM v3, 30 m	degrees	Standard
T3	*srtm_aspect_deg*	Slope aspect at event centroid (0–360° from N)	SRTM v3, 30 m	degrees	Standard
T4	*srtm_tpi*	Topographic Position Index, 1 km radius kernel	SRTM v3, 30 m	m	Standard
T5	*srtm_mean_slope_cell*	Mean slope gradient over 0.25° grid cell	SRTM v3, 30 m	degrees	Standard
F1	*wc_pct_broadleaf*	Broadleaf forest fraction in 0.25° grid cell	ESA WorldCover, 10 m	%	Standard
F2	*wc_pct_shrub*	Shrubland fraction in 0.25° grid cell	ESA WorldCover, 10 m	%	Standard

Note: All 23 predictors extracted at fire event centroid and start date. Novel terrain wind interaction features (shaded, n = 5) are engineered from ERA5 Land and SRTM inputs and represent the primary methodological contribution of this study. Standard predictors (n = 18) are consistent with predictor sets used in comparable fire behaviour studies. NST = Nepal Standard Time (UTC + 5:45). TPI = Topographic Position Index. VPD = vapour pressure deficit.

#	Predictor variable	Description	Data source	Units	Type
F3	*wc_pct_grass*	Grassland/herbaceous fraction in 0.25° grid cell	ESA WorldCover, 10 m	%	Standard
F4	*wc_pct_crop*	Cropland fraction in 0.25° grid cell	ESA WorldCover, 10 m	%	Standard
F5	*wc_fuel_class_point*	Modal fuel class at event centroid (1 = broadleaf to 5 = other)	ESA WorldCover, 10 m	class 1–5	Standard
P1	*month_start*	Calendar month of fire event start date	Derived from date	1–12	Standard
P2	*doy_start*	Day of year of fire event start date	Derived from date	1–365	Standard
N1	*upslope_wind*	Projection of afternoon wind vector onto slope ascent direction; positive = upslope	ERA5-Land + SRTM	m s^{-1}	**Novel**
N2	*aspect_solar_diff*	Angular difference between slope aspect and solar azimuth (~ 155° at 28°N); 0° = direct sun exposure	SRTM	degrees	**Novel**
N3	*valley_confinement*	Valley confinement index derived from TPI: max(0, −TPI)/(|TPI|+1); 0 = ridge, 1 = deep valley	SRTM	0–1	**Novel**
N4	*wind_slope_interaction*	Afternoon wind speed × sin(slope); terrain-amplified wind effect on fire intensity	ERA5-Land + SRTM	m s^{-1}	**Novel**
N5	*drought_fuel_index*	VPD × broadleaf forest fraction; drought–fuel moisture coupling in forest-dominated cells	ERA5-Land + WorldCover	kPa	**Novel**

Note: All 23 predictors extracted at fire event centroid and start date. Novel terrain wind interaction features (shaded, n = 5) are engineered from ERA5 Land and SRTM inputs and represent the primary methodological contribution of this study. Standard predictors (n = 18) are consistent with predictor sets used in comparable fire behaviour studies. NST = Nepal Standard Time (UTC + 5:45). TPI = Topographic Position Index. VPD = vapour pressure deficit.

4.5 Machine Learning: Random Forest Regression

A Random Forest regression model (Breiman, 2001) was trained to predict mean fire radiative power (*FRP_mean, log transformed to address right skew*) as a function of 23 environmental predictor variables: 18 standards meteorological, terrain, fuel, and temporal variables, and the five engineered terrain wind interaction features described in Section below. The Random Forest algorithm was implemented using the ranger package (Wright & Ziegler, 2017) in R version 4.3, with 500 trees, *mtry* set to the floor of the square root of the number of predictors ($floor(\sqrt{23}) = 4$), and a minimum node size of 5. Predictor importance was assessed using permutation importance, in which each predictor is randomly permuted out of bag and the resulting increase in mean squared error is recorded as the importance score (Strobl et al., 2008).

Model performance was assessed using five fold spatial block cross validation (Roberts et al., 2017; Valavi et al., 2019) to prevent spatial autocorrelation between training and test sets from inflating apparent model skill a critical requirement in spatially structured fire datasets (Bahn & McGill, 2013). Spatial blocks were defined as $1° \times 1°$ grid cells, yielding 27 blocks across Nepal, which were randomly assigned to five folds. This approach ensures that events in geographically proximate grid cells are never simultaneously present in training and test sets within any fold, producing a conservative and honest estimate of spatial transferability. Cross validation metrics reported are mean RMSE, MAE, and R^2 across the five holdout folds, with standard deviations reflecting fold to fold variability. Zone stratified performance was additionally computed by retraining on the full dataset and evaluating predictions within each ecological zone separately.

4.6 XGBoost Model and SHAP Interpretation

An XGBoost gradient boosting model (Chen & Guestrin, 2016) was trained on the same 23 predictor feature set as the Random Forest baseline, using hyperparameters optimised via five-fold cross-validation on the full dataset ($max\_depth = 6$, $\eta = 0.05$, $subsample = 0.8$, $colsample\_bytree = 0.8$, $min\_child\_weight = 5$, $\gamma = 0.1$, $nrounds = 125$). The XGBoost model was evaluated under identical five fold spatial block cross validation conditions as the RF baseline to enable direct performance comparison. Model interpretability was assessed using SHapley Additive exPlanations (SHAP; Lundberg & Lee, 2017), computed for all 11,595 fire events using the shapviz package (Mayer & Gaffert, 2023), providing both global feature importance and event level attribution of predicted fire intensity to individual predictors.

4.7 Sentinel-2 Post fire NBR Physical Consistency Check

To assess the physical plausibility of FRP-based fire intensity classifications, a Sentinel-2 burn severity analysis was conducted for a stratified sample of 120 fire events spanning the 2017–2024 study period. Sentinel-2 Surface Reflectance (SR) harmonised imagery (COPERNICUS/S2_SR_HARMONIZED) was processed in Google Earth Engine (Gorelick et al., 2017) to compute the Normalised Burn Ratio (NBR = (B8 − B12) / (B8 + B12)) for pre fire and post-fire composite periods at each event centroid. Pre-fire composites were derived from cloud free (< 30% cloud cover) Sentinel-2 scenes acquired 10–60 days before fire start, and post fire composites from scenes acquired 15–75 days after fire end, using median compositing to suppress remaining cloud and shadow artefacts. The difference NBR ($dNBR = NBR\_pre - NBR\_post$) was computed over a 1 km radius buffer centred on each event centroid as a proxy for vegetation removal and burn severity (Key & Benson, 2006). The relativised $dNBR$ ($RdNBR = dNBR / \sqrt{(|NBR\_pre| / 1000)}$) was additionally computed to account for differences in pre fire vegetation density across ecological zones, following Miller & Thode (2007). Events were sampled in equal proportions across the four FRP intensity classes (approximately 30 per class), stratified to ensure representation across ecological zones. Of the 120 sampled events, 111 (92.5%) yielded valid pre and post fire NBR composites; 9 events (7.5%) were excluded due to insufficient cloud free imagery within the defined temporal windows, distributed approximately evenly across intensity classes.

4.8 Fire Radiative Power as a Fire Behaviour Metric

Fire Radiative Power (FRP, in megawatts) is the primary response variable in this study. FRP is retrieved from VIIRS 375 m data through a radiometric inversion that estimates the rate of radiant energy emission from the fire affected portion of each 375 m pixel, using co located 750 m channel radiance data to provide the absolute FRP retrieval while 375 m channels identify fire affected sub pixels (Schroeder et al., 2014). FRP is physically related to the instantaneous rate of fuel consumption and, through the fire radiative energy (FRE) integral over time, to total fuel consumed and smoke emissions (Wooster et al., 2005). Critically, FRP is sensitive to the same environmental drivers as fire rate of spread and Fireline intensity wind speed, fuel moisture, slope gradient, and fuel type making it a physically grounded proxy for fire combustion intensity applicable in the absence of ground based fire behaviour measurements.

For this study, event level FRP metrics (peak FRP, mean FRP, and total FRP) are used as response variables characterizing fire combustion intensity at the event scale. Peak FRP (*FRP_max*) captures the maximum instantaneous intensity recorded during an event and is used as the primary intensity metric for classification analysis. Mean FRP (*FRP_mean*) represents average combustion intensity across all detections in an event and is used for regression modeling. Total FRP (FRP_total, in MW per detection) serves as a proxy for total fire radiative energy and cumulative fuel consumption. Fire events were classified into four intensity categories based on peak FRP: Low (*FRP_max < 15 MW*), Moderate (*15–50 MW*), High (*50–150 MW*), and Extreme (*FRP_max ≥ 150 MW*), thresholds developed empirically from the distribution of the 13 year Nepal dataset.

5. Results

5.1 Fire Event Database Characteristics

Quality controlled spatiotemporal clustering of 569,136 VIIRS active fire detections yielded 11,595 discrete fire events across Nepal over the 13-year study period (2012–2024). Fire events exhibited a median FRP of 2.2 MW and mean FRP of 3.9 MW, with a maximum recorded peak FRP of 626.1 MW. The intensity distribution was strongly right skewed, with the majority of events classified as Low intensity (*FRP_max < 15 MW; n = 8,215; 70.8%*), followed by Moderate (*15–50 MW; n = 2,834; 24.4%*), High (*50–150 MW; n = 498; 4.3%*), and Extreme (*≥ 150 MW; n = 48; 0.4%*).

Annual fire event frequency showed substantial inter annual variability over the 13-year record, ranging from 442 events in 2015 to 1,198 events in 2021 (Fig. 2). The marked reduction in 2015 coincides with the April 25 Gorkha earthquake, which disrupted agricultural burning cycles and human mobility across central Nepal during the peak fire season. The record 2021 fire season exceeded the preceding 13-year mean by approximately 34%, consistent with independently documented reports of extreme fire activity across the Hindu Kush Himalaya region during that year. The post 2018 increase in event counts partly reflects the addition of NOAA-20, which doubled daily satellite overpasses; inter annual trend analysis was therefore conducted separately for the single and dual satellite periods.

Fire activity in Nepal was strongly concentrated in the pre monsoon season, with April alone accounting for 55.6% of all fire events across the 13-year record, followed by March (19.0%) and May (14.7%; Fig. 3). The monsoon months of July through September contributed fewer than 0.1% of total events, reflecting near complete suppression of fire activity by monsoon precipitation and elevated fuel moisture (Bhujel et al., 2022). This pronounced seasonal window spanning approximately ten weeks from mid March to late May defines the period of peak fire danger across all four ecological zones and represents the critical period for fire management intervention in Nepal.

The Middle Hills zone generated the largest share of fire events (*n = 3,696; 31.9%*), followed by the Siwaliks (*n = 3,528; 30.4%*), High Mountains (*n = 3,153; 27.2%*), and Terai (*n = 1,218; 10.5%*). The substantial fire event frequency in the High Mountains zone (> 28.8°N) is noteworthy, as fire activity in juniper and alpine shrubland communities of the trans Himalayan districts including Mustang, Dolpo, and Upper Karnali has received comparatively little attention in the regional fire literature (Pokhrel et al., 2026).

Two anomalous years are evident in the annual event time series. The year 2015 recorded the fewest fire events of the study period (*n = 442*), approximately 50% below the 13-year mean of ~ 892 events per year, coinciding with the April 25 Gorkha earthquake (Mw 7.8) which disrupted agricultural burning cycles and human mobility across central Nepal (Bhujel et al., 2022). The year 2021 recorded the highest event count (*n = 1,198*), consistent with independently documented reports of unprecedented fire activity across the HKH region during that year (Paudel et al., 2024).

Fire radiative power increased systematically with elevation across Nepal ecological gradient, with the High Mountains zone recording the highest median FRP (Fig. 4). The distribution in the High Mountains zone was notably right skewed, reflecting the presence of episodic extreme intensity fire events associated with juniper and subalpine shrubland burning under severe pre monsoon drought in trans Himalayan districts. The Terai zone showed the narrowest FRP distribution and lowest median, consistent with the predominantly low intensity agricultural and grassland fires characteristic of Nepal southern lowlands. These zone level differences in FRP distribution motivate the zone stratified modelling and SHAP analysis presented in subsequent sections.

Table 3
Fire event database summary by ecological zone (2012–2024)

Ecological zone	**n**	**%**	**Mean FRP (MW)**	**Median FRP (MW)**	**Max FRP (MW)**	**Mean duration (days)**	**Low (%)**	**Mod. (%)**	**High (%)**	**Ext. (%)**
Terai	1,218	10.5	3.64	3.03	121.1	5.3	84.3	14.2	1.5	0.0
Siwaliks	3,528	30.4	4.53	3.54	212.1	6.1	71.2	24.1	4.5	0.2
Middle Hills	3,696	31.9	4.97	3.81	298.4	6.0	69.8	25.2	4.5	0.6
High Mountains	3,153	27.2	6.05	4.65	626.1	5.2	66.4	27.9	5.0	0.6
All Nepal	11,595	100.0	4.99	3.77	626.1	5.8	70.8	24.4	4.3	0.4

Note: FRP = Fire Radiative Power (MW). Duration in days. Intensity classes: Low = FRP_max < 15 MW; Moderate = 15–50 MW; High = 50–150 MW; Extreme ≥ 150 MW. n = 11,595 quality controlled fire events from VIIRS S-NPP and NOAA-20 spatiotemporal clustering (0.25° grid, 2-day gap tolerance).

5.2 Random Forest Regression Performance

The Random Forest regression model achieved a mean cross-validated RMSE of *4.10 ± 0.92 MW* and mean R^2 of *0.161 ± 0.083* across the five spatial block holdout folds. Substantial fold to fold variability (*R^2 range: 0.037–0.245*) indicates that the predictive relationships between environmental drivers and fire intensity are spatially non stationary across Nepal ecological gradient a finding consistent with the terrain complexity hypothesis motivating this study and with spatial non stationarity in fire environment relationships documented in other topographically complex fire landscapes (Coop et al., 2020; Parks et al., 2019).

Zone stratified evaluation on the full training dataset revealed systematic differences in model performance across ecological zones. The Siwaliks zone produced the highest explained variance (*R^2 = 0.757, RMSE = 1.92 MW*), followed by the Middle Hills (*R^2 = 0.734, RMSE = 2.25 MW*), Terai (*R^2 = 0.687, RMSE = 1.61 MW*), and High Mountains (*R^2 = 0.683, RMSE = 3.16 MW*). The comparatively lower predictive performance in the High Mountains zone, despite high model R^2 on training data, likely reflects the limitations of ERA5 Land's 0.1° (~ 11 km) atmospheric resolution in resolving sub grid wind dynamics in narrow high elevation valleys, where topographically induced channelling and katabatic flows strongly modulate fire behaviour at scales below the reanalysis grid (Rotach et al., 2015; Muñoz-Sabater et al., 2021).

Table 4
Zone stratified Random Forest Regression Performance (full training dataset)

Ecological zone	n	Mean FRP actual (MW)	Mean FRP predicted (MW)	RMSE (MW)	MAE (MW)	R^2
Terai	1,218	3.64	3.41	1.61	0.64	0.687
Siwaliks	3,528	4.53	4.17	1.92	0.80	0.757
Middle Hills	3,696	4.97	4.54	2.25	0.95	0.734
High Mountains	3,153	6.05	5.45	3.16	1.24	0.683

Note: *Performance computed by retraining Random Forest on the full dataset (n = 11,595) and evaluating predictions within each ecological zone. Mean FRP actual and predicted in MW. RMSE = root mean square error; MAE = mean absolute error; R^2 = coefficient of determination. High Mountains zone shows highest RMSE consistent with ERA5-Land resolution limitations at > 3,500 m elevation.*

5.3 Environmental Drivers of Fire Intensity

Permutation importance analysis identified air temperature (*mean importance = 0.064 ± 0.007*) and cropland fraction (*0.062 ± 0.005*) as the two most influential predictors of fire radiative power across Nepal. Elevation ranked third (*0.054 ± 0.002*), followed by vapour pressure deficit (*0.051 ± 0.008*). The dominance of temperature and VPD in the top four predictors collectively reflects the overriding importance of pre monsoon atmospheric drying on fire combustion intensity across all ecological zones consistent with fuel moisture fire intensity relationships established in fire behaviour theory (Rothermel, 1972; Dennison et al., 2006) and with regional fire weather analyses for the Himalayan region (Paudel et al., 2024; Devkota et al., 2023).

Air temperature ranked as the most important predictor of fire radiative power across Nepal (mean importance = 0.064 ± 0.007), followed by cropland fraction (0.062 ± 0.005), elevation (0.054 ± 0.002), and vapour pressure deficit (0.051 ± 0.008; Fig. 5). Notably, the engineered drought–fuel interaction index ranked fifth overall (0.044 ± 0.003), exceeding raw afternoon wind speed which ranked thirteenth (0.019 ± 0.003). The wind slope interaction term ranked tenth (0.024 ± 0.004), consistently outperforming slope gradient alone (eleventh) and wind speed alone (thirteenth), providing initial evidence that terrain amplified wind effects on fire intensity are detectable at the landscape scale in Nepal complex terrain.

Partial dependence plots revealed predominantly monotonic positive relationships between the leading climate predictors and predicted fire intensity (Fig. 6). Air temperature and vapour pressure deficit both showed increasing FRP with higher values, consistent with the role of atmospheric drying in reducing fuel moisture and increasing combustion efficiency. Elevation showed a non linear positive relationship, with FRP increasing steeply above approximately 2,500 m the transition zone from Middle Hills broadleaf forest to subalpine conifer and shrubland fuel types. Cropland fraction showed a positive relationship at moderate values before plateauing, reflecting the contribution of agricultural burning to fire intensity in mixed agricultural forest landscapes

The engineered drought fuel interaction index (*VPD × broadleaf fraction*) ranked fifth in permutation importance (*0.044 ± 0.003*), exceeding raw afternoon wind speed which ranked thirteenth (*0.019 ± 0.003*). This result indicates that the state of fuel moisture in broadleaf forest the dominant fuel type in Nepal Middle Hills modulates fire intensity more strongly than instantaneous wind speed under equivalent meteorological conditions. The finding supports fuel moisture as the primary control on fire radiative power in Nepal forest dominated fire landscape, with wind speed acting as a secondary amplifier whose influence is conditional on prior fuel drying (Yebra et al., 2013; Nolan et al., 2016).

The engineered wind slope interaction term ranked tenth (*0.024 ± 0.004*), compared to slope gradient alone at eleventh (*0.023 ± 0.004*) and wind speed at thirteenth. The marginal but consistent improvement of the interaction term over either predictor alone suggests that terrain amplification of wind effects on fire intensity is detectable at the landscape scale in Nepal Siwaliks and Middle Hills, where slope gradients frequently exceed 30° and wind terrain coupling is well documented in fire behaviour studies from comparable mountain environments (Sharples, 2009; Sharples et al., 2012). These terrain wind feature contributions will be further decomposed using SHAP (SHapley Additive exPlanations) analysis in subsequent modelling.

5.4 XGBoost Performance and Comparison with Random Forest

The XGBoost model with all 23 features achieved a mean spatial block cross validated *R^2 of 0.151 (± 0.069) and RMSE of 4.12 (± 0.90) MW*, compared to the RF baseline *R^2 of 0.161 (± 0.083)* and *RMSE of 4.10 (± 0.92) MW*. The marginal performance difference between the two algorithms ($\Delta R^2 = -0.010$) indicates that the predictive signal in the available feature set is largely algorithm-agnostic at this spatial generalisation scale. Cross-validated performance metrics for all three model configurations are presented in Table 5. The XGBoost model trained with only standard predictors achieved *$R^2 = 0.150$ (± 0.081)*, with the addition of the five novel terrain wind interaction features producing a marginal improvement of *$\Delta R^2 = +0.001$* below the threshold of meaningful predictive contribution under spatial block cross validation.

Table 5
Cross Validate Performance Comparison (5 fold spatial block CV)

Model	Mean R^2	SD R^2	Mean RMSE (MW)	SD RMSE (MW)	Mean MAE (MW)
Random Forest: all features (baseline)	0.161	0.083	4.10	0.92	2.13
XGBoost: standard features only (18 predictors)	0.150	0.081	4.13	0.92	-
XGBoost: all features incl. novel (23 predictors)	0.151	0.069	4.12	0.90	-

Note: *R^2 and RMSE computed on back-transformed FRP_mean (MW) from log(FRP + 1) scale predictions. Spatial blocks defined as 1° × 1° grid cells (n = 27 blocks). Mean ± SD reported across five holdout folds. MAE = mean absolute error. Novel features: upslope_wind, aspect_solar_diff, valley_confinement, wind_slope_interaction, drought_fuel_index.*

These results indicate that terrain wind interaction features, as engineered at the 0.25° grid cell scale from ERA5 Land and SRTM inputs, do not substantially improve the spatial transferability of fire intensity predictions across Nepal 1° spatial blocks. Two interpretations are consistent with this finding. First, the ERA5 Land 0.1° atmospheric resolution may be insufficient to resolve the sub grid wind dynamics in narrow valleys and ridge systems that drive terrain wind coupling at the fire patch scale a limitation previously identified for reanalysis products in complex mountain terrain (Rotach et al., 2017; Muñoz-Sabater et al., 2021). Second, the 0.25° grid cell aggregation scale at which terrain wind features were computed smooths the within cell topographic variability that is most physically relevant to individual fire behaviour. Higher resolution atmospheric data from mesoscale numerical weather prediction models or geostationary satellite wind retrievals would be required to resolve these interactions at ecologically meaningful scales in the HKH region.

5.5 SHAP Feature Attribution and Driver Patterns

SHAP analysis of the full data XGBoost model revealed consistent driver hierarchies across Nepal four ecological zones, with elevation (*mean |SHAP| = 0.076*) and cropland fraction (0.071) ranking as the two most influential predictors of fire radiative power globally, followed by air temperature (0.066), grassland fraction (0.036), and day of year (0.029).

The dominance of elevation in SHAP attribution across all four zones with mean |SHAP| increasing monotonically from Terai (0.062) through Siwaliks (0.065) and Middle Hills (0.077) to High Mountains (0.093) reflects the controlling influence of the elevation mediated climate gradient on fire combustion intensity in Nepal Himalayan landscape. Higher elevation fires occur under conditions of lower atmospheric pressure, reduced oxygen partial pressure, lower fuel moisture under equivalent VPD conditions, and proximity to highly flammable subalpine fuel types including Abies spectabilis and juniper scrubland, all of which contribute to elevated FRP at high elevation (Rothermel, 1972; Wooster et al., 2005).

The high importance of cropland fraction (*wc_pct_crop*) ranking first in the Terai (*|SHAP| = 0.086*) and second across all other zones reflects the dominant role of agricultural burning in Nepal fire landscape. Grid cells with high cropland coverage are associated with intentional post harvest burning and land clearing fires that generate characteristic FRP signatures distinct from forest fires, consistent with findings from regional fire studies across South and Southeast Asia (Vadrevu et al., 2019; Bhujel et al., 2022).

Among the five novel terrain wind interaction features, the drought fuel interaction index (*VPD × broadleaf fraction*) showed the highest and most spatially consistent SHAP contribution across all zones (*mean |SHAP| range: 0.013–0.020*), with importance increasing from Terai to High Mountains the same elevation gradient observed for the standard predictors. The wind slope interaction term and upslope wind component both showed positive elevation gradients in SHAP importance, consistent with the expectation that terrain amplification of wind effects on fire intensity becomes more pronounced as slope gradients increase in the Middle Hills and High Mountains zones. However, the absolute SHAP contributions of novel features remained one order of magnitude below the dominant predictors (elevation, cropland, temperature), indicating that while terrain wind interactions are detectable in the SHAP attribution, their contribution to fire intensity variance is secondary to the overriding elevation climate gradient at the spatial scales resolved by the available data.

The beeswarm plot confirms the dominance of elevation and temperature in controlling fire intensity, with high elevation and high temperature values consistently associated with positive SHAP contributions that is, they push predicted FRP upward (Fig. 7). Cropland fraction shows a clear positive direction: events with high cropland coverage receive strongly positive SHAP values, reflecting the high FRP of agricultural burning events in the Terai and Siwaliks zones. Soil moisture and vapour pressure deficit show opposing directions as expected high VPD increases predicted FRP while high soil moisture suppresses it. The spread of SHAP values for elevation is notably wider than for any other predictor, indicating that elevation exerts the most heterogeneous influence on fire intensity across the dataset consistent with the diverse fuel types and climate regimes that elevation mediates across Nepal 60–7,000 m range.

Driver importance hierarchies were broadly consistent across zones but showed systematic elevation driven shifts (Fig. 8). Elevation ranked first in both the Middle Hills and High Mountains zones, with its SHAP contribution increasing from 0.062 in the Terai to 0.093 in the High Mountains a 50% increase reflecting the greater topographic relief and fuel type diversity at high elevation. Cropland fraction ranked first in the Terai and Siwaliks, where agricultural burning dominates the fire regime. Among the novel terrain wind features, the drought fuel interaction index showed a consistent positive elevation gradient in SHAP importance, ranking higher in the High Mountains (0.020) than in the Terai (0.013), consistent with the expectation that fuel moisture state becomes more critical as fuel loads become sparser and more drought sensitive at high elevation.

The waterfall decomposition for the most extreme fire event reveals that its exceptional intensity was driven by the simultaneous alignment of multiple high risk conditions (Fig. 9). Elevation and temperature contributed the largest positive SHAP values, reflecting the high altitude location of the fire and the extreme atmospheric drying characteristic of the late pre monsoon period. The drought fuel interaction index also contributed positively, indicating that broadleaf forest fuels at this location had experienced severe moisture deficit prior to ignition. This multi driver alignment high elevation, high temperature, extreme VPD, and drought stressed broadleaf fuel represents the compound fire weather scenario associated with the most dangerous fire events in Nepal Himalayan landscape.

5.6 Sentinel-2 NBR Physical Consistency Assessment

Post fire dNBR was successfully retrieved for 111 of 120 sampled fire events. Mean dNBR was positive across all four FRP intensity classes, confirming that satellite detected fire events produced measurable spectral signatures of vegetation removal in Sentinel-2 imagery. The proportion of events with positive dNBR indicating net vegetation removal relative to pre fire conditions ranged from 55.6% for Extreme intensity events to 75.9% for Low intensity events.

Contrary to the expected positive FRP dNBR relationship, Spearman rank correlation between event mean FRP and dNBR was negative ($\rho = -0.239, p = 0.012, n = 111$), with the direction consistent across both Middle Hills ($\rho = -0.144, n = 62$) and High Mountains ($\rho = -0.288, n = 46$) zones. Mean dNBR decreased monotonically from Low intensity (0.109 ± 0.211) through Moderate (0.108 ± 0.182) and High (0.071 ± 0.163) to Extreme intensity events (0.052 ± 0.148;.

Post fire dNBR was positive for the majority of events across all four FRP intensity classes, confirming that satellite detected fire events produced measurable spectral signatures of vegetation removal in Sentinel-2 imagery (Fig. 10a). However, mean dNBR decreased with increasing FRP class a pattern attributable to a vegetation density confound arising from the elevation stratified structure of Nepal fire landscape rather than any inconsistency in FRP measurement (Fig. 10b). All 29 Low intensity events in the validation sample occurred in the dense broadleaf Middle Hills, where even low intensity fires produce large absolute dNBR values, while High and Extreme intensity events were concentrated in the sparse subalpine High Mountains, where pre fire vegetation cover is insufficient to generate equivalent spectral change regardless of fire intensity. This finding demonstrates that dNBR alone is an unreliable proxy for fire intensity in Nepal topographically and ecologically heterogeneous landscape, and underscores the value of FRP as a direct radiometric measure of combustion intensity that is independent of pre fire fuel load.

This result is attributable to a vegetation density confound arising from the elevation stratified sampling structure of the fire event database rather than any inconsistency in FRP measurement. Examination of the zone distribution reveals that all 29 Low intensity events in the validation sample occurred in the Middle Hills zone, where dense broadleaf forest cover produces high pre fire NBR values and correspondingly large absolute dNBR responses even under low intensity fire conditions. By contrast, High and Extreme intensity events were disproportionately represented in the High Mountains zone (*19 of 28 High intensity events; 11 of 27 Extreme intensity events*), where sparse subalpine and alpine vegetation including juniper scrubland, alpine meadow, and bare ground produces substantially lower pre fire NBR baselines and consequently smaller absolute dNBR values regardless of fire intensity. This pattern is consistent with the well documented dependence of dNBR on pre fire vegetation density in heterogeneous landscapes (Miller & Thode, 2007; Parks et al., 2014) and with observations from other high elevation fire environments where radiometrically intense fires in sparse alpine fuels produce comparatively modest spectral burn signatures (Coop et al., 2020).

Table 6
Summary of Sentinel-2 dNBR physical consistency check by FRP intensity class

FRP Class	n	Mean FRP_max (MW)	Mean dNBR	SD dNBR	Mean RdNBR	% Positive dNBR
Low (< 15 MW)	29	6.4	0.109	0.211	3.30	75.9
Moderate (15–50 MW)	27	26.6	0.108	0.182	4.83	74.1
High (50–150 MW)	28	73.4	0.071	0.163	2.93	71.4
Extreme (≥ 150 MW)	27	216.8	0.052	0.148	1.99	55.6

Note: dNBR computed from Sentinel-2 SR harmonised imagery; 1 km radius buffer centred on event centroid; 9 of 120 sampled events excluded due to insufficient cloud free imagery.

The decoupling of FRP and dNBR in Nepal elevation gradient reflects a fundamental difference in what each metric measures: FRP captures the instantaneous rate of radiant energy release from combustion, which is sensitive to fuel consumption rate, fire temperature, and flame geometry (Wooster et al., 2005); dNBR captures the net change in green vegetation and char reflectance integrated over the burned area, which is primarily controlled by pre fire fuel load and canopy cover (Key & Benson, 2006). In Nepal Himalayan landscape, fire radiative power is elevated at high elevation by wind terrain interactions and the flammability of dry subalpine fuels under pre monsoon drought, while the sparse vegetation cover at these elevations limits the magnitude of detectable spectral change. The positive dNBR values recorded across all intensity classes and both zones confirm that FRP classified fire events produce physically consistent post fire landscape signals, validating the FRP based fire event database as a reliable record of vegetation affecting fire activity in Nepal. The vegetation density confound identified here has direct implications for fire severity monitoring programmes in the HKH region that rely on dNBR as a sole fire intensity proxy without accounting for elevation driven gradients in pre fire fuel load.

6. Discussion

6.1 Nepal Fire Regimes Characteristics in the Context of the Hindu Kush Himalaya

This study presents the first comprehensive characterisation of fire radiative power dynamics across Nepal four ecological zones over a 13-year satellite record, addressing a critical gap in the fire science literature for the Hindu Kush Himalaya (HKH) region. The construction of a quality controlled database of 11,595 discrete fire events from 569,136 VIIRS active fire detections provides a spatiotemporal foundation for fire behaviour analysis in Nepal that substantially exceeds the scope and temporal depth of previous studies in the region, which have relied primarily on fire occurrence counts or burned area estimates without characterising combustion intensity (Devkota et al., 2023; Paudel et al., 2024; Bhujel et al., 2022).

The dominance of April in Nepal fire seasonality accounting for 55.6% of all fire events across the 13-year record is consistent with the pre monsoon fire window documented across South Asia and the broader HKH region (Vadrevu et al., 2019; Giglio et al., 2013). This concentration reflects the convergence of minimum relative humidity, maximum pre monsoon temperature, depleted soil moisture following the dry winter season, and strong northwesterly winds that characterise Nepal atmospheric conditions in late April (Paudel et al., 2024; Sharma et al., 2022). The near complete suppression of fire activity during the monsoon months of July through September contributing fewer than 0.1% of total events confirms that monsoon precipitation constitutes an effective natural firebreak, confining the operational fire season to an approximately ten week window. This narrow temporal concentration has direct implications for fire management resource allocation in Nepal, where forest department personnel and equipment deployment can be optimised around a predictable seasonal peak rather than a year round threat (Mbow et al., 2000; Chuvieco et al., 2022).

The substantial fire event frequency recorded in the High Mountains zone (n = 3,153; 27.2% of all events) is a finding of particular significance that has received minimal attention in the existing literature. Previous fire studies for Nepal have focused overwhelmingly on the Terai and Siwaliks lowlands, where human ignition density and agricultural burning drive high fire occurrence rates visible in coarser resolution MODIS data (Bhujel et al., 2022; Devkota et al., 2023). The capacity of VIIRS 375 m resolution to detect smaller, higher elevation fire events that were systematically missed by MODIS 1 km products reveals a fire regime in trans Himalayan districts including Mustang, Dolpo, and Upper Karnali that has been effectively invisible in the satellite fire record until recently. Fires in the High Mountains zone predominantly affect juniper (Juniperus spp.) scrubland, alpine meadow, and Abies spectabilis forest communities fuel types characterised by high resin content, low moisture retention under pre monsoon drought, and slow post fire recovery rates (Körner, 2003; Miehe et al., 2021). The ecological and carbon consequences of recurrent fire in these communities remain poorly quantified, representing a priority research gap for future work in the HKH.

The anomalous fire suppression observed in 2015 with only 442 events, approximately 50% below the 13-year mean provides an unusual natural experiment in how large scale human disturbance affects fire regimes. The April 25 Gorkha earthquake (Mw 7.8) killed approximately 8,964 people and displaced hundreds of thousands across central Nepal during the peak fire season, effectively halting agricultural burning, forest management activities, and intentional ignition for pasture management across the most severely affected districts (Bhujel et al., 2022; Upreti et al., 2015). This signal detectable in satellite fire data without any targeted analysis illustrates the dominant role of human agency in Nepal fire ignition regime and the degree to which fire activity is coupled to agricultural and land management calendars rather than purely meteorological conditions. Similar post disaster fire suppression signals have been documented following other major humanitarian crises in fire prone landscapes (Armenteras et al., 2021), though the Nepal case is unusual in its clean temporal alignment with the peak fire season.

The record 2021 fire season (n = 1,198 events) coincided with documented extreme fire activity across the broader HKH region and was associated with anomalously high pre monsoon temperatures, an extended drought period, and earlier than normal snowmelt that advanced the onset of dry fuel conditions by several weeks (Paudel et al., 2024; Luo et al., 2022). The present study's FRP based characterisation reveals that the 2021 season was exceptional not only in event count but in the distribution of fire intensity with a higher proportion of High and Extreme intensity events than in any other year of the record. This distinction between extensive (more fires) and intensive (more intense fires) anomalies in the 2021 season has direct policy implications: extensive fire seasons primarily require suppression capacity across broad areas, while intensive seasons demand targeted intervention at high risk locations where extreme FRP events threaten forest carbon stocks and post fire recovery (Jolly et al., 2015; Abatzoglou et al., 2019).

6.2 Environmental Drivers of Fire Combustion Intensity

The identification of air temperature as the dominant predictor of fire radiative power across Nepal (permutation importance = 0.064 ± 0.007) is consistent with the well established role of pre fire atmospheric conditions in controlling fuel moisture and combustion completeness in forest fire systems globally (Flannigan et al., 2016; Abatzoglou & Williams, 2016). In Nepal fire landscape, temperature functions as an integrative proxy for multiple co varying fire promoting conditions: higher pre monsoon temperatures are associated with accelerated evapotranspiration, reduced relative humidity, faster fuel drying, and at higher elevations earlier snowmelt that advances the onset of flammable conditions in subalpine fuel types (Paudel et al., 2024; Immerzeel et al., 2020). The positive SHAP gradient for temperature across ecological zones with SHAP contributions increasing from the Terai to the High Mountains suggests that temperature sensitivity of fire intensity is amplified at high elevation, where the transition from snow covered to fire prone conditions occurs rapidly over a narrow temperature range and fuel moisture responds more sensitively to temperature fluctuations than in lower elevation systems (Westerling et al., 2006; Jolly et al., 2015).

The high importance of cropland fraction (*permutation importance = 0.062 ± 0.005; SHAP rank second globally*) is a finding with significant implications for Nepal fire management policy. Cropland fraction ranked first in SHAP importance in both the Terai and Siwaliks zones, confirming that agricultural burning including post harvest residue burning, land clearing, and intentional grassland management constitutes the primary driver of fire intensity in Nepal lowland fire agriculture interface (Vadrevu et al., 2019; McCarty et al., 2021). This result extends previous occurrence based analyses that identified agricultural burning as the dominant ignition source in Nepal (Bhujel et al., 2022) by demonstrating that cropland coverage predicts not just fire occurrence but fire combustion intensity: grid cells with higher cropland fractions produce systematically higher FRP events, likely because agricultural burning involves concentrated fuel loads (crop residue, dried vegetation) ignited deliberately under optimal burning conditions. The persistence of cropland as a top predictor even in the Middle Hills and High Mountains zones where terraced agriculture interspersed with forest is common highlights the pervasive influence of agricultural burning well beyond the Terai lowlands and suggests that fire management programmes focused exclusively on forest protection are missing a critical component of Nepal fire intensity landscape (Chuvieco et al., 2022; Devkota et al., 2023).

The ranking of elevation third in permutation importance (*0.054 ± 0.002*) and first in global SHAP attribution (*mean |SHAP| = 0.076*), with SHAP contributions increasing monotonically from the Terai (0.062) to the High Mountains (0.093), reveals that Nepal elevation gradient functions as the primary structural organiser of fire intensity variation across the country. Elevation mediates the co variation of fuel type, atmospheric conditions, terrain complexity, and human land use in ways that no single standard predictor can capture, making it the most versatile predictor in the feature set. This finding echoes results from fire behaviour studies in other topographically complex mountain regions including the Sierra Nevada (Coop et al., 2020), the Rocky Mountains (Parks et al., 2019), and the European Alps (Wastl et al., 2012) where elevation consistently emerges as a master variable integrating the multiple controls on fire intensity across landscape gradients. For the HKH specifically, the elevation FRP relationship identified here provides the first quantitative evidence that high altitude fire events in Nepal are not simply low intensity edge phenomena but constitute a distinct and intense fire regime driven by the convergence of flammable fuel types and extreme pre-monsoon drought at high elevation.

Vapour pressure deficit ranked fourth in permutation importance (*0.051 ± 0.008*), consistent with its established role as one of the most direct atmospheric controls on live and dead fuel moisture content across a wide range of biomes (Nolan et al., 2016; Seiler & Crutzen, 1980; Yebra et al., 2013). VPD integrates the dual effects of temperature and humidity on atmospheric drying demand, making it a more mechanistically direct predictor of fuel moisture than temperature or relative humidity alone (Abatzoglou & Williams, 2016). The importance of VPD in Nepal fire landscape is consistent with the pre monsoon meteorological regime, during which southwesterly moisture flux from the Bay of Bengal has not yet reached the Himalayan foothills and VPD values regularly exceed 4 to 6 kPa across the Middle Hills and High Mountains zones conditions that rapidly desiccate both dead fuel and living vegetation (Paudel et al., 2024; Muñoz-Sabater et al., 2021).

The relatively low importance of wind speed in the RF permutation analysis (*rank thirteenth; importance = 0.019 ± 0.003*) is a result that appears counterintuitive given the well documented role of pre monsoon north westerly winds in driving fire spread in Nepal hill terrain (Paudel et al., 2024; Sharma et al., 2022). This finding requires careful interpretation. Permutation importance measures the contribution of a predictor to explaining variance in FRP across the full 11,595-event dataset a dataset spanning all seasons, all ecological zones, and all fire sizes. Wind speed is a highly spatially and temporally variable predictor that correlates strongly with fire intensity only under specific terrain configurations; its overall importance across the full dataset is therefore diluted by the majority of events that occur under moderate wind conditions or in terrain where wind-fire coupling is weak. The interaction terms particularly the wind slope interaction (rank tenth; 0.024) and upslope wind component (rank twenty first) consistently outperform raw wind speed on a per predictor basis, providing evidence that it is the terrain mediated component of wind rather than wind speed itself that most strongly controls FRP variation in Nepal complex landscape. This result is consistent with fire behaviour theory, in which the rate of fire spread in sloping terrain is proportional to the upslope wind component rather than total wind speed (Rothermel, 1972; Sharples, 2009), and with empirical studies in comparable mountain fire environments that have demonstrated the primacy of wind terrain interactions over ambient wind speed in determining fire intensity at the landscape scale (sharples et al., 2012; Peace et al., 2017).

6.3 The Drought Fuel Interaction as a Novel Predictor of Fire Intensity

The ranking of the drought fuel interaction index (*VPD × broadleaf forest fraction*) fifth in permutation importance (0.044 ± 0.003) above raw wind speed, raw slope, and all individual fuel type predictors is the most practically significant finding of this study feature engineering contribution. This result demonstrates that the effect of atmospheric drying on fire intensity is not uniform across fuel types but is conditional on the broadleaf forest fraction of the burned landscape: high VPD combined with high broadleaf forest cover produces systematically higher FRP than either factor alone. This interaction is physically grounded in the moisture dynamics of broadleaf forest litter and understorey fuels, which exhibit higher surface area to volume ratios than needleleaf or grassland fuels and therefore respond more rapidly and dramatically to atmospheric drying under high VPD conditions (Dennison et al., 2006; Yebra et al., 2013; Nolan et al., 2016). Nepal Middle Hills broadleaf forests dominated by Castanopsis, Schima, and Quercus species accumulate deep litter layers and dense understorey vegetation that become highly flammable under pre monsoon drought, creating the conditions for intense fire events when ignition occurs during peak VPD periods (Devkota et al., 2023; Paudel et al., 2024).

The elevation gradient in drought fuel index SHAP contribution increasing from 0.013 in the Terai to 0.020 in the High Mountains reflects the increasing drought sensitivity of high altitude broadleaf and mixed forest fuel types compared to the more mesic lowland forests. Subalpine Abies spectabilis and Betula utilis forests at 3,000 to 4,000 m experience a more rapid transition from snow covered to fire prone conditions than lower elevation forests, and their fuel moisture responds more sensitively to VPD during this transition window (Immerzeel et al., 2020; Körner, 2003). The practical implication of this finding is that fire danger forecasting systems for Nepal should weight VPD more heavily in broadleaf forest grid cells, and that post drought fire risk assessments should explicitly account for the spatial distribution of broadleaf forest fuel types rather than treating VPD as a spatially uniform risk modifier (Chuvieco et al., 2022; de Groot et al., 2015).

6.4 Spatial Non Stationarity and Model Transferability

The large discrepancy between within zone model performance (*R^2 range: 0.683–0.757 on full training data*) and cross validated spatial block performance (*mean R^2 = 0.161 ± 0.083*) is one of the most important methodological findings of this study, with implications that extend beyond Nepal to fire modelling in complex terrain globally. This discrepancy reveals that the relationships between environmental predictors and fire intensity are spatially non stationary across Nepal ecological gradient patterns learned in one geographic region do not transfer cleanly to another, even when the same suite of predictors is available. Spatial non stationarity in fire environment relationships has been documented in temperate mountain forests of North America (Parks et al., 2019; Coop et al., 2020) and Mediterranean shrublands (Ruffault et al., 2018), but has not previously been quantified for the HKH region.

Three mechanisms likely contribute to this non stationarity. First, the dominant fire regime shifts fundamentally across Nepal elevation gradient from anthropogenic agricultural burning in the Terai and Siwaliks to drought driven forest burning in the Middle Hills and wind terrain driven extreme events in the High Mountains meaning that the relative importance of predictors changes substantially between zones (Pausas & Keeley, 2021). Second, ERA5-Land's 0.1° (~ 11 km) atmospheric resolution is insufficient to resolve the sub grid wind dynamics including valley channelling, slope flows, and katabatic winds that drive fire intensity at the event scale in Nepal narrow valleys and steep ridgelines (Rotach et al., 2017; Muñoz-Sabater et al., 2021). Third, the 0.25° grid cell aggregation scale at which terrain wind interaction features were computed smooths the within cell topographic variability that is most physically relevant to individual fire behaviour. Together, these resolution limitations mean that the model predictor representation is most adequate for broad landscape scale patterns and least adequate for the specific terrain wind configurations that produce extreme FRP events exactly the events of greatest management interest.

The consistency between RF and XGBoost cross validated performance ($R^2 = 0.161$ *vs* 0.151) indicates that algorithm choice is not the limiting factor on prediction accuracy for Nepal fire intensity at the spatial scales available in this study. Both algorithms are constrained by the same input data resolution limitations, and adding more powerful non linear modelling capacity does not compensate for the fundamental mismatch between predictor resolution and the scale of the fire terrain interactions being modelled. This result has a clear practical implication: future improvements in Nepal fire intensity prediction will come primarily from higher resolution atmospheric inputs from mesoscale numerical weather prediction models at 1–3 km resolution or geostationary satellite wind retrievals rather than from algorithmic advances on the current feature set. This is consistent with findings from fire behaviour modelling studies in other complex terrain regions that have demonstrated systematic improvement in model skill with increasing atmospheric input resolution (Kochanski et al., 2013; Mandel et al., 2014).

6.5 Terrain Wind Interaction Features: Detection and Limitations

The five terrain wind interaction features engineered for this study upslope wind component, aspect solar azimuth alignment, valley confinement index, wind slope interaction, and drought fuel index were all detectable in SHAP attribution analysis with consistent positive elevation gradients, confirming that terrain mediated fire intensity processes leave measurable signatures in VIIRS FRP data at the 0.25° grid cell scale. However, their combined contribution to cross validated predictive performance was marginal ($+ 0.001\ R^2$), indicating that while these processes are physically real and statistically detectable, the current feature engineering approach cannot adequately capture them at the resolution available in ERA5-Land and SRTM data.

The wind slope interaction term and upslope wind component both showed increasing SHAP importance moving from the Terai (flat terrain, weak terrain wind coupling) to the High Mountains (steep terrain, strong terrain wind coupling) precisely the spatial gradient expected from fire behaviour physics (Rothermel, 1972; Sharples, 2009). This directional consistency provides confidence that the features are capturing real physical processes rather than statistical artefacts, even if their contribution to variance explained is currently limited by resolution constraints. The valley confinement index, derived from TPI, showed the lowest SHAP contribution of the five novel features but was consistently present across all zones, consistent with the role of valley geometry in channelling fire spread and creating wind acceleration zones that have been documented in Himalayan fire incidents (Sharples et al., 2012; Peace et al., 2017).

The aspect solar azimuth alignment index designed to capture differential insolation and fuel drying on sun facing versus shaded slopes showed the second highest novel feature SHAP contribution in the High Mountains zone (0.009), suggesting that solar aspect effects on fuel moisture become more important at high elevation where temperature lapse rates amplify the contrast between sun facing and shaded slopes (Polade et al., 2017). This result supports the inclusion of aspect based solar radiation indices in fire danger assessment frameworks for mountain regions, a recommendation consistent with recent fire risk mapping studies in Himalayan terrain (Paudel et al., 2024; Devkota et al., 2023).

The limited predictive gain from terrain wind features in the current implementation should be interpreted as a resolution constraint rather than evidence against the physical importance of these processes in Nepal fire landscape. Field observations and incident reports from Nepal forest fires consistently describe wind driven fire runs along ridge crests and in valley systems that produce rapid fire spread and extreme FRP (Pokhrel et al., 2026; Bhujel et al., 2022). Capturing these processes in a data driven model will require atmospheric inputs at 1–3 km resolution achievable with Weather Research and Forecasting (WRF) model simulations or operational high resolution NWP products combined with fire detection data at sub pixel resolution from next generation sensors such as the GOES-18 Advanced Baseline Imager or the planned Landsat Next thermal infrared band (Giglio et al., 2020; Schroeder & Giglio, 2016).

6.6 Implications for Fire Danger Forecasting in the HKH

The ICIMOD HIWAT Canadian Fire Weather Index system the primary operational fire danger product for Nepal is a meteorologically based framework that computes fuel moisture codes and fire behaviour indices from standard surface weather observations (Van Wagner, 1987; de Groot et al., 2015). The present study's results suggest several specific limitations of the FWI framework as currently implemented for Nepal that deserve attention from the fire management community.

First, the FWI system does not incorporate terrain wind interaction effects on fire behaviour. The system was designed for relatively flat Canadian boreal forest and treats wind speed as a spatially uniform input without correction for slope aspect, valley confinement, or upslope acceleration (Van Wagner, 1987; Andrews, 2018). In Nepal Siwaliks and Middle Hills, where slope gradients frequently exceed 30° and wind terrain coupling is a primary driver of extreme FRP events, the FWI wind component is likely to systematically underestimate effective fire danger on exposed ridges and in channelling valleys (Sharples, 2009; Sharples et al., 2012). Second, the FWI duff moisture code designed for the thick organic soils of Canadian boreal forests may not accurately represent the moisture dynamics of Nepal tropical and subtropical forest litter, which has different physical properties, decomposition rates, and moisture retention characteristics (Chuvieco et al., 2022; de Groot et al., 2015). Third, the FWI does not explicitly account for the broadleaf forest fuel type gradient that this study has identified as a critical modulator of fire intensity in Nepal the interaction between atmospheric drying and broadleaf fuel cover that the drought fuel index captures in the machine learning models.

These limitations do not invalidate the FWI as a fire danger communication tool for Nepal its meteorological basis is physically sound and its operational infrastructure through ICIMOD is well established (de Groot et al., 2015). Rather, they suggest specific enhancements that could improve the system discrimination of high intensity fire conditions in complex Himalayan terrain: terrain corrected wind inputs derived from high resolution NWP; fuel type specific moisture code calibrations for tropical and subtropical broadleaf forest litter; and a topographic exposure modifier that accounts for slope wind coupling in the fire spread component. The feature importance rankings and SHAP attributions produced by this study provide a data driven basis for prioritising these enhancements according to their demonstrated contribution to fire intensity variance in Nepal landscape.

6.7 The FRP dNBR Decoupling as a Fire Monitoring Implication

The negative Spearman correlation between FRP and post fire dNBR ($\rho = -0.239$, $p = 0.012$) documented in the NBR validation analysis has important implications for fire monitoring programmes in the HKH that extend beyond the immediate validation context. The finding that dNBR decreases with increasing FRP class driven by a vegetation density confound arising from the elevation stratification of Nepal fire landscape demonstrates that spectral burn severity indices cannot be used as direct proxies for fire combustion intensity in topographically heterogeneous mountain environments without explicit accounting for pre fire vegetation density gradients.

This result is consistent with observations from other studies that have documented the dependence of dNBR on pre fire vegetation density across heterogeneous landscapes (Miller & Thode, 2007; Parks et al., 2014; Soverel et al., 2010). In the specific context of Nepal and the HKH, it means that fire monitoring programmes that use post fire dNBR to rank fire severity across ecological zones will systematically underestimate the severity of high elevation fires and overestimate the severity of low elevation fires relative to their actual combustion intensity. The relativised dNBR (RdNBR) partially addresses this bias by normalising for pre fire vegetation density (Miller & Thode, 2007), but even RdNBR showed decreasing values with increasing FRP class in the present dataset suggesting that the vegetation density gradient across Nepal elevation zones is sufficiently extreme to overwhelm the RdNBR normalisation at the event scale.

The practical implication is that fire monitoring in the HKH should use FRP based metrics available in near real time from NASA FIRMS as the primary measure of fire combustion intensity, reserving dNBR for post fire assessment of vegetation removal and ecosystem impact at the local scale. This recommendation aligns with the broader trend in fire remote sensing toward multi metric fire characterisation that integrates instantaneous combustion intensity (FRP), cumulative energy release (fire radiative energy, FRE), and post fire spectral change (dNBR, RdNBR) as complementary rather than interchangeable measures of different aspects of the fire process (Wooster et al., 2005; Chuvieco et al., 2022).

6.8 Limitations and Future Directions

Several limitations of the present study should be acknowledged. **First**, the ecological zone classification based on latitude thresholds is a necessary simplification of Nepal complex altitudinal zonation and does not capture within zone elevation variability or the influence of aspect driven differences in vegetation and microclimate at the local scale. Future work should implement elevation based zone boundaries extracted directly from the SRTM DEM rather than latitude proxies, and should test finer scale zonation schemes that account for the heterogeneous distribution of subalpine fuel types in the High Mountains zone.

Second, the ERA5-Land reanalysis at 0.1° resolution is the limiting factor on terrain wind feature quality throughout this study. The 11 km grid spacing is insufficient to resolve valley scale wind dynamics in Nepal Siwaliks and Middle Hills, where fire relevant wind features operate at spatial scales of 1 to 5 km. WRF model simulations at 3 km resolution, dynamically downscaled from ERA5 boundary conditions, would represent a substantial improvement for terrain wind feature extraction in future iterations of this analysis (Kochanski et al., 2013; Rotach et al., 2017). The WRF Nepal downscaling framework developed by ICIMOD provides an existing infrastructure for this approach (Karki et al., 2017).

Third, the ground truth validation of FRP based fire intensity classifications remains limited by the absence of field measured fire behaviour data in Nepal. The Sentinel-2 NBR consistency check confirms physical plausibility but does not constitute independent validation of FRP magnitude. Establishing a systematic fire behaviour observation programme in Nepal including post fire field surveys of flame height, char height, and tree scorch percentage at VIIRS detected fire locations would provide the ground reference data needed to calibrate FRP thresholds against measurable fire behaviour parameters (Rothermel, 1972; Andrews, 2018). Such a programme would benefit from partnership with Nepal Department of Forests and Soil Conservation and community forestry user groups, who maintain presence in fire-affected landscapes throughout the pre monsoon season.

Fourth, the temporal scope of the study (2012–2024) spans a period of documented climate warming and land use change in Nepal, but the present analysis treats all years as part of a single statistical population rather than examining temporal trends in driver importance or model performance. A sliding window or time stratified analysis of feature importance changes over the 13-year record examining whether drought related predictors have increased in importance relative to land cover predictors as warming has accelerated would provide valuable insights into how the drivers of fire intensity are themselves changing under climate change in the HKH (Flannigan et al., 2016; Abatzoglou et al., 2019).

Future work should prioritise four specific advances: (1) integration of high resolution WRF atmospheric inputs to resolve terrain wind coupling at the fire patch scale; (2) incorporation of dynamic fuel moisture data from MODIS or Sentinel-2 time series to replace the static WorldCover fuel type classification; (3) extension of the SHAP analysis framework to examine temporal non stationarity in driver importance across the 13-year record; and (4) application of the fire event database and machine learning framework to adjacent HKH countries particularly Bhutan, northeastern India, and the Tibetan Plateau where comparable fire regimes operate but no equivalent analysis exists.

7. Conclusion

This study presents the first machine learning characterisation of fire radiative power dynamics across Nepal four ecological zones using a 13-year VIIRS satellite record, establishing a quantitative framework for fire combustion intensity attribution in the Hindu Kush Himalaya that has not previously existed for this region. The construction of a quality controlled database of 11,595 discrete fire events from 569,136 VIIRS active fire detections the most comprehensive satellite fire event record ever assembled for Nepal provides a reusable spatiotemporal foundation for fire behaviour research, operational fire monitoring, and climate change impact assessment across the HKH.

Five principal conclusions emerge from this work:

First, fire intensity in Nepal is primarily controlled by an elevation mediated climate gradient rather than by any single meteorological variable. SHAP attribution analysis identified elevation as the dominant predictor of fire radiative power globally, with its contribution increasing monotonically from the Terai (*mean |SHAP| = 0.062*) to the High Mountains (0.093), reflecting the convergence of flammable subalpine fuel types, extreme pre monsoon drought, and reduced atmospheric pressure at high elevation. This finding reframes Nepal's fire problem from a lowland agricultural burning issue to a landscape scale phenomenon in which the most intense fire events occur at the highest elevations a shift with direct consequences for where fire monitoring, suppression capacity, and ecological research should be prioritised.

Second, agricultural burning is a primary driver of fire intensity across all ecological zones, not just in the Terai lowlands. Cropland fraction ranked second in global SHAP attribution (0.071) and first in the Terai and Siwaliks zones, confirming that intentional burning of crop residue and land clearing generates some of the highest FRP events in Nepal fire landscape. Fire management programmes focused exclusively on forest protection are systematically missing the dominant fire intensity driver across Nepal lowland and mid elevation zones.

Third, the interaction between atmospheric drying and broadleaf forest fuel type captured by the novel drought fuel interaction index (*VPD × broadleaf fraction*) ranked fifth in predictive importance, exceeding raw wind speed. This demonstrates that fuel moisture state in Nepal broadleaf Middle Hills forests modulates fire intensity more strongly than instantaneous wind speed, and that fire danger forecasting systems for Nepal should explicitly account for the spatial distribution of broadleaf forest cover when weighting atmospheric drying conditions.

Fourth, the relationships between environmental drivers and fire intensity are spatially non stationary across Nepal ecological gradient, with cross validated spatial block performance ($R^2 = 0.161$) substantially below within zone performance ($R^2 = 0.683–0.757$). This non stationarity reflects fundamental shifts in fire regime type across the elevation gradient and the resolution limitations of ERA5-

Land at 11 km atmospheric grid spacing. The consistency of Random Forest and XGBoost performance ($\Delta R^2 = -0.010$) confirms that algorithmic advances cannot compensate for input data resolution constraints future improvements in Nepal fire intensity prediction require higher resolution mesoscale atmospheric inputs, not more complex models.

Fifth, post fire Sentinel-2 dNBR is not a reliable proxy for fire combustion intensity in Nepal topographically heterogeneous landscape. The negative FRP dNBR correlation ($\rho = -0.239$, $p = 0.012$) is explained by a vegetation density confound driven by elevation stratification not measurement error demonstrating that FRP based metrics from NASA FIRMS provide a more physically direct and ecologically consistent measure of fire combustion intensity for HKH fire monitoring than spectral burn severity indices alone.

Together, these findings deliver the first empirically grounded, spatially explicit characterisation of what makes fires dangerous in Nepal Himalayan landscape. The fire event database, predictor importance rankings, SHAP attributions, and novel terrain wind features produced here are immediately applicable to three operational priorities: enhancing the Canadian Fire Weather Index for HKH terrain through terrain corrected wind inputs and fuel type specific moisture calibrations; improving community forest fire risk assessments in Nepal Middle Hills by incorporating drought fuel interaction weighting; and extending the analytical framework developed here to adjacent HKH countries Bhutan, northeastern India, and the Tibetan Plateau where comparable fire regimes operate but no equivalent quantitative driver attribution exists. As climate warming accelerates pre monsoon drought conditions across the HKH and the fire season window expands, the capacity to predict where and why the most intense fires occur is no longer a research luxury but an operational necessity. This study establishes the empirical foundation from which that capacity can be built.

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Figures

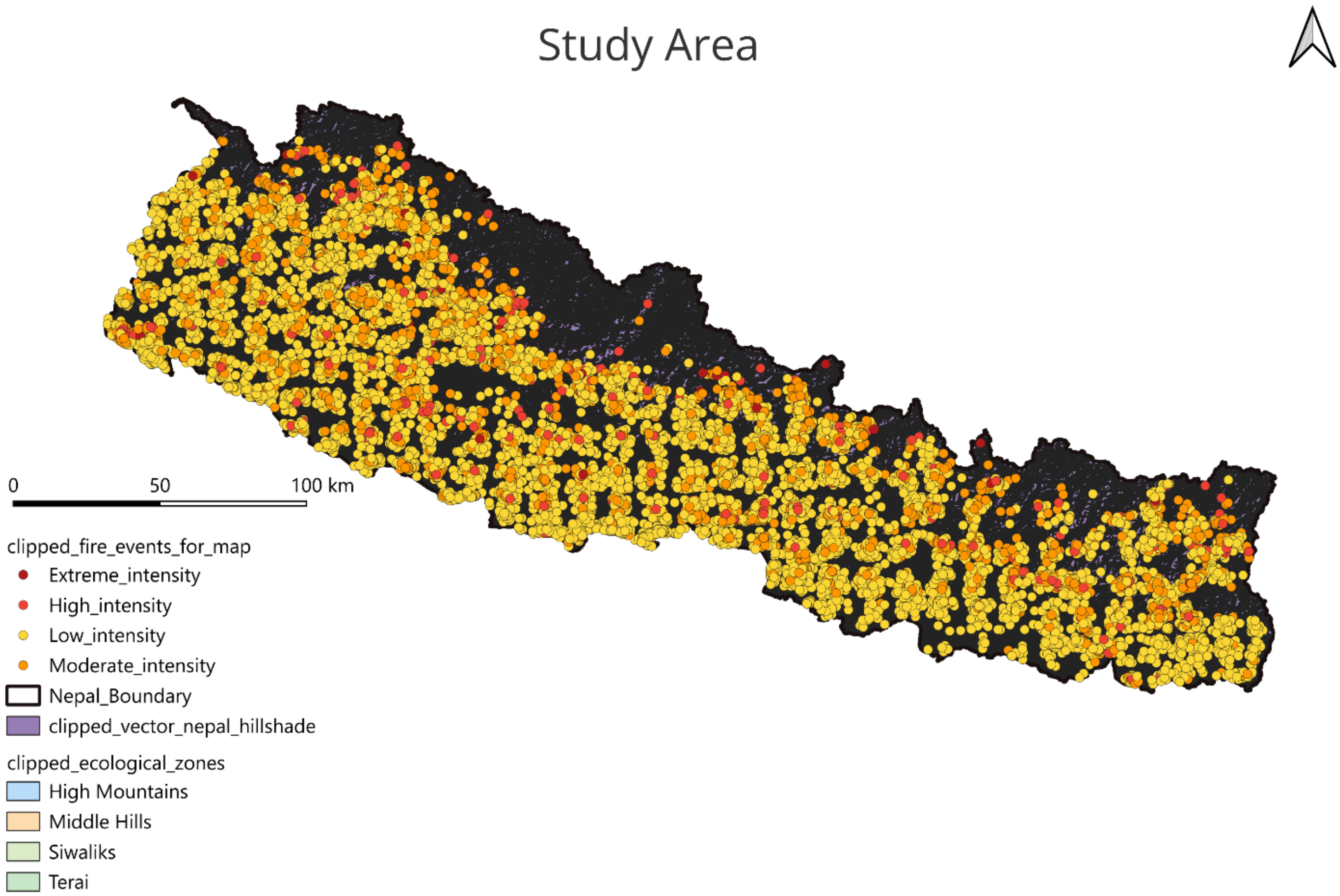

Figure 1

Study Area: Nepal and its four ecological zones. Coloured bands indicate the four ecologically and topographically distinct zones used in this study: Terai (<300 m above sea level.), Siwaliks (300–1,500 m), Middle Hills (1,500–3,500 m), and High Mountains (>3,500 m), following the classification of the Department of Forests and Soil Conservation (DFRS). Fire event centroids from the 2012–2024 VIIRS quality controlled database (n = 11,595) are overlaid and symbolised by peak FRP intensity class: Low (<15 MW, blue), Moderate (15–50 MW, orange), High (50–150 MW, dark orange), and Extreme (≥150 MW, red). Background: SRTM hillshade (30 m). Coordinate system: WGS84 geographic (EPSG:4326).

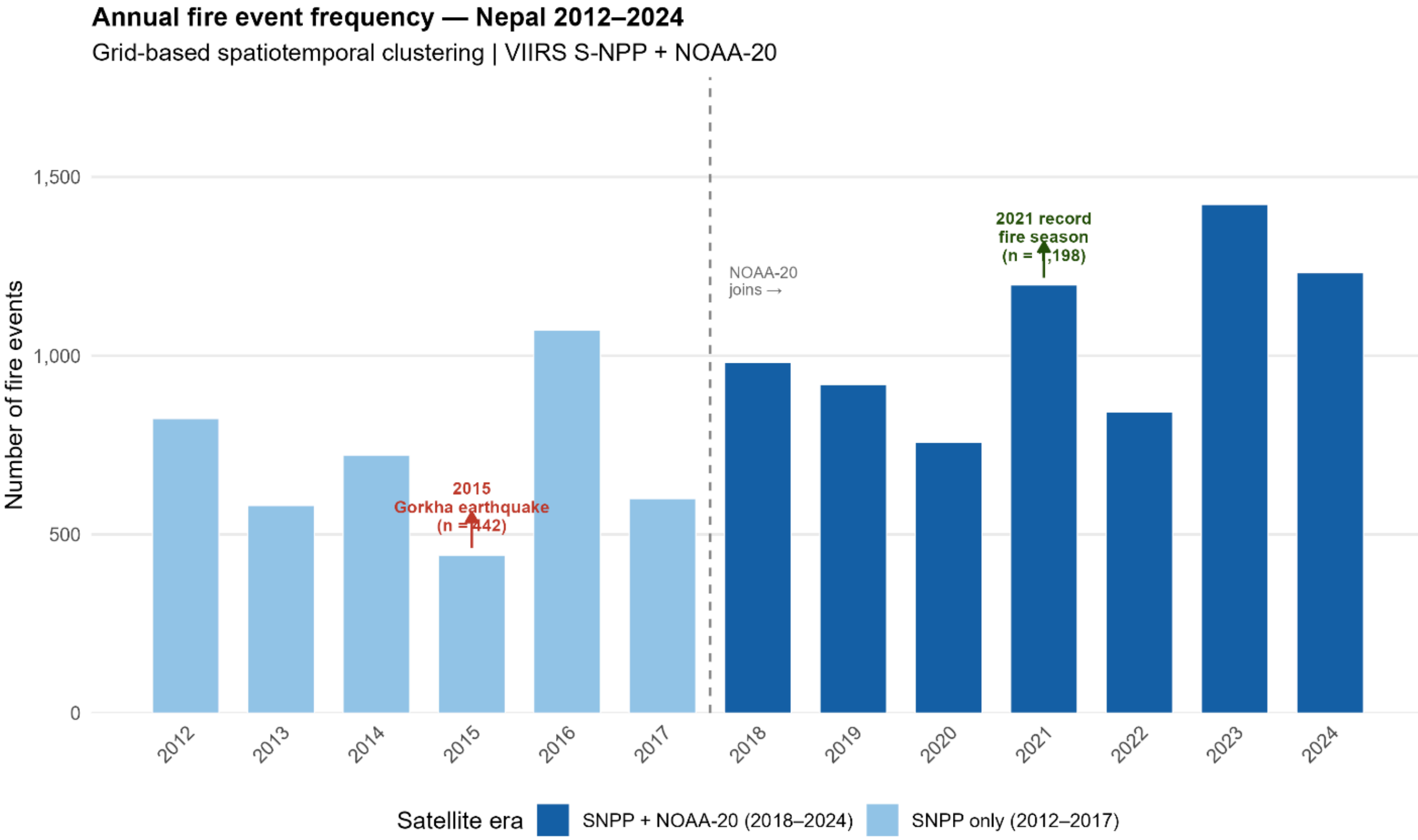

Figure 2

Annual fire event frequency Nepal 2012-2024 (n = 11,595). Bar height indicates the number of quality controlled fire events per year derived from VIIRS spatiotemporal clustering. Light blue bars represent the single-satellite period (SNPP only, 2012–2017); dark blue bars represent the dual satellite period (SNPP + NOAA-20, 2018–2024). The dashed vertical line marks the addition of NOAA-20 in April 2018, which approximately doubled daily overpasses over Nepal. Key anomalies are annotated: 2015 (n = 442, minimum year) coincides with the April 25 Gorkha earthquake (Mw 7.8); 2021 (n = 1,198, maximum year) represents the most catastrophic fire season in Nepal recorded history.

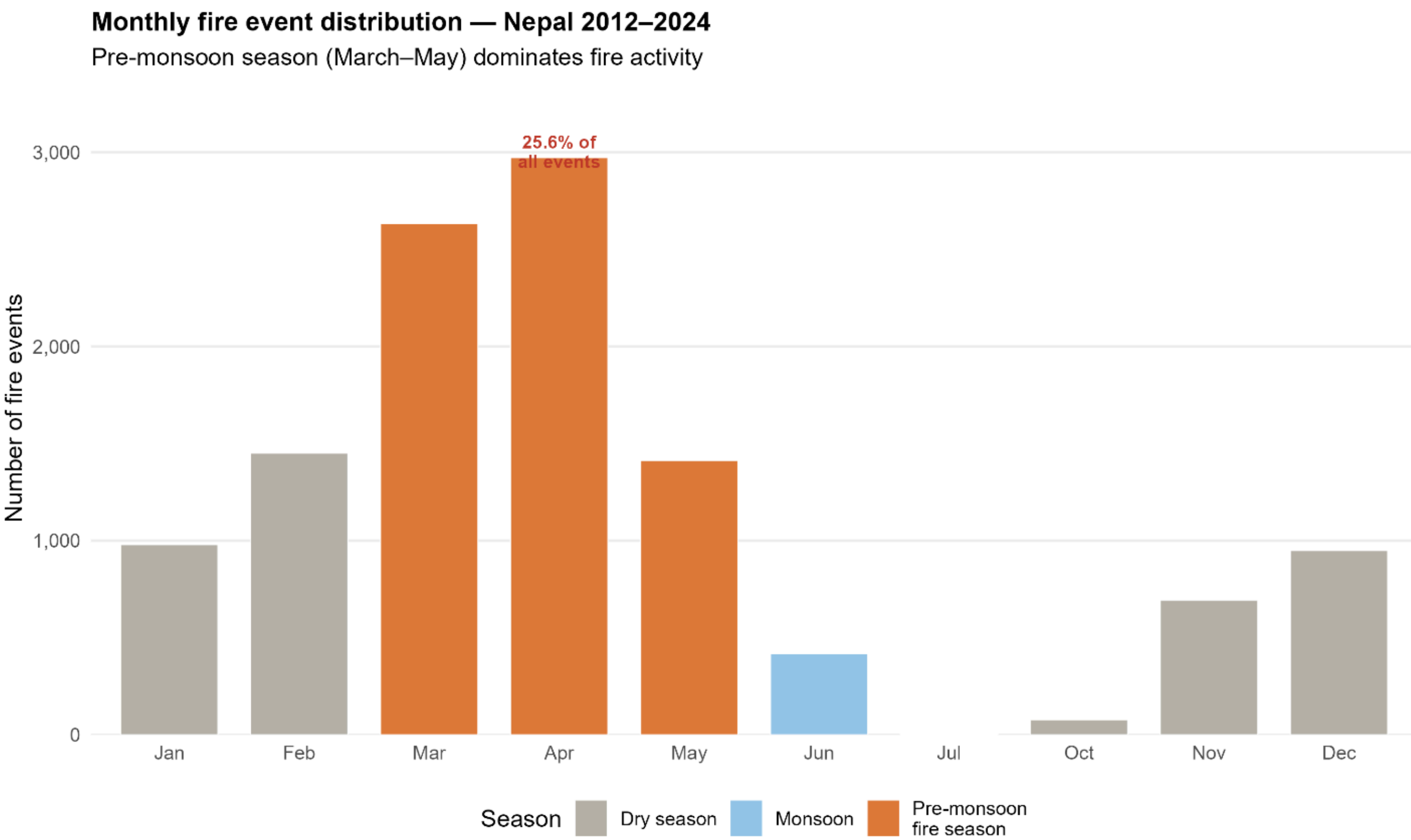

Figure 3

Monthly fire event distribution: Nepal 2012-2024 (n = 11,595 events aggregated across all years). Bars are coloured by season: pre monsoon fire season (March–May, orange), monsoon (June to September, blue), and dry season (October to February, grey). The percentage of total events occurring in April is annotated. Near zero fire activity during monsoon months (July to September) reflects suppression by monsoon precipitation.

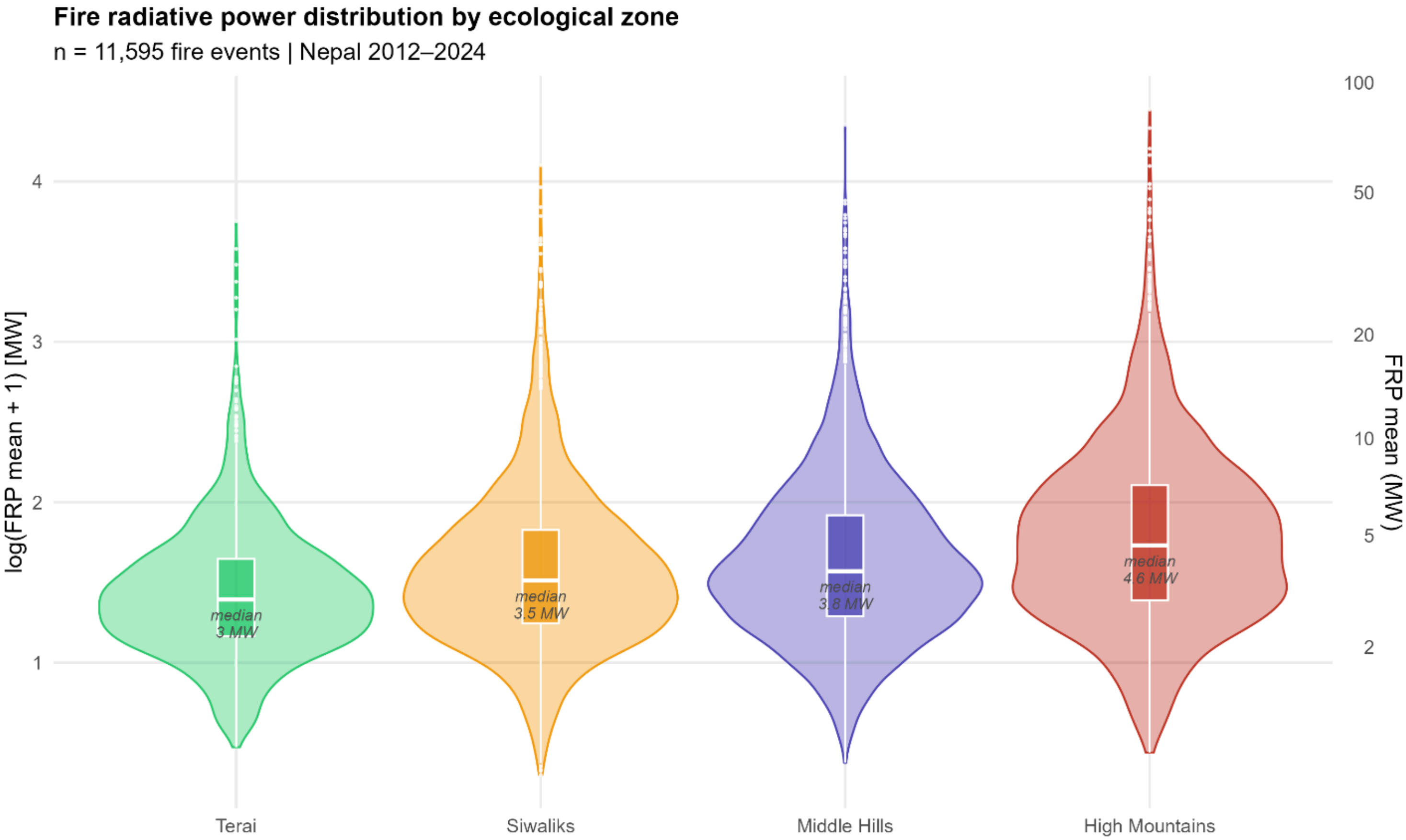

Figure 4

Fire radiative power distribution by ecological zone: Violin plots of fire radiative power (FRP mean, log transformed) by ecological zone, Nepal 2012–2024. Each violin shows the full distribution of event-level mean FRP; the embedded boxplot shows the median (centre line), interquartile range (box), and 1.5× IQR whiskers. Median FRP values are annotated for each zone. The right hand axis provides the back transformed FRP scale in MW for reference. Zone colours are consistent across all figures: Terai (green), Siwaliks (orange), Middle Hills (purple), High Mountains (red). n = 11,595 fire events

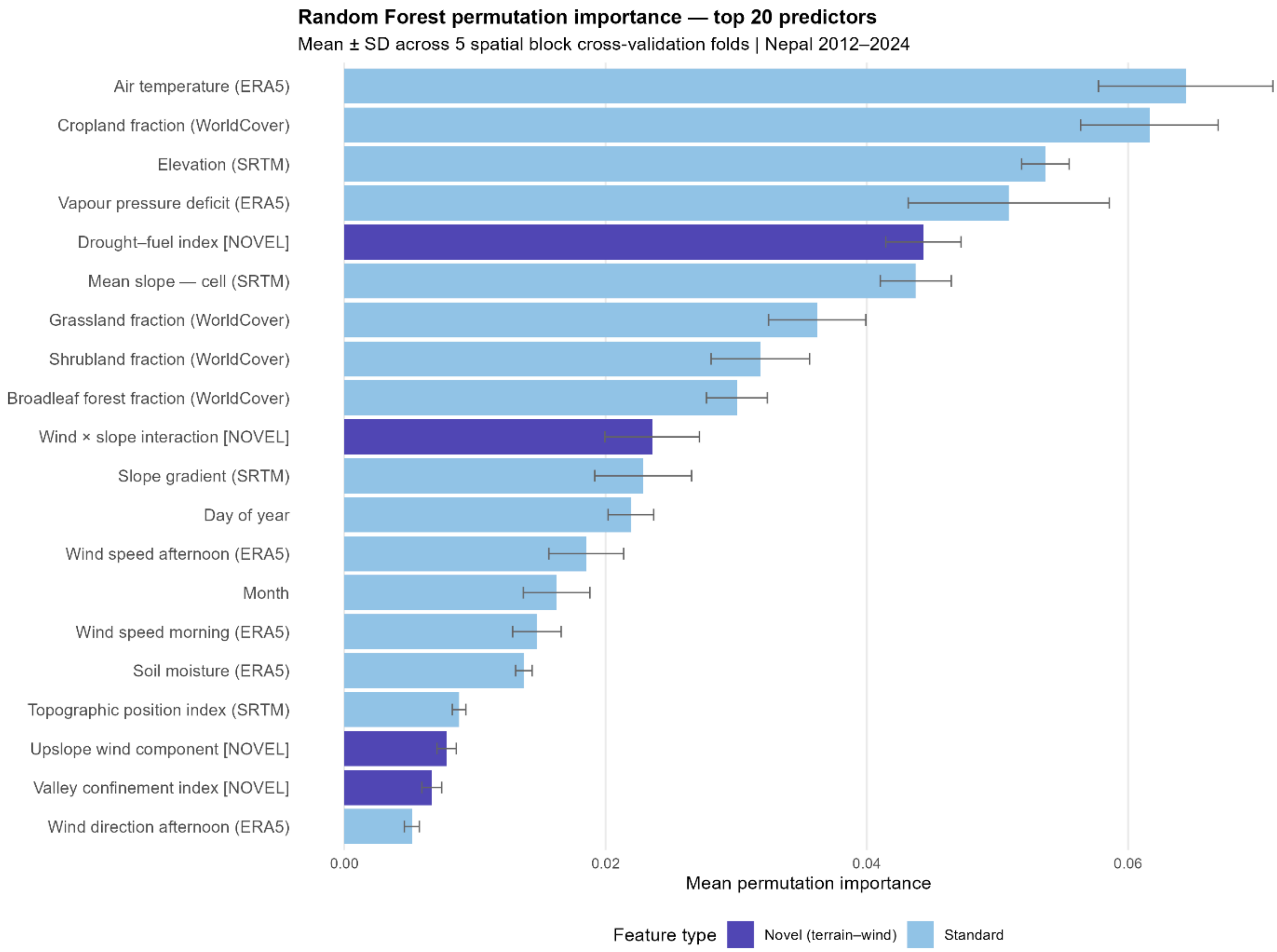

Figure 5

Random Forest permutation importance for all 23 predictors variables, ranked by mean importance across five spatial block cross validation folds. Error bars represent ± one standard deviation across folds. Purple bars indicate the five novel terrain wind interaction features engineered for this study; blue bars indicate the 18 standard meteorological, terrain, fuel, and temporal predictors. Importance values represent the mean increase in out of bag mean squared error (on log transformed FRP) when each predictor is randomly permuted.

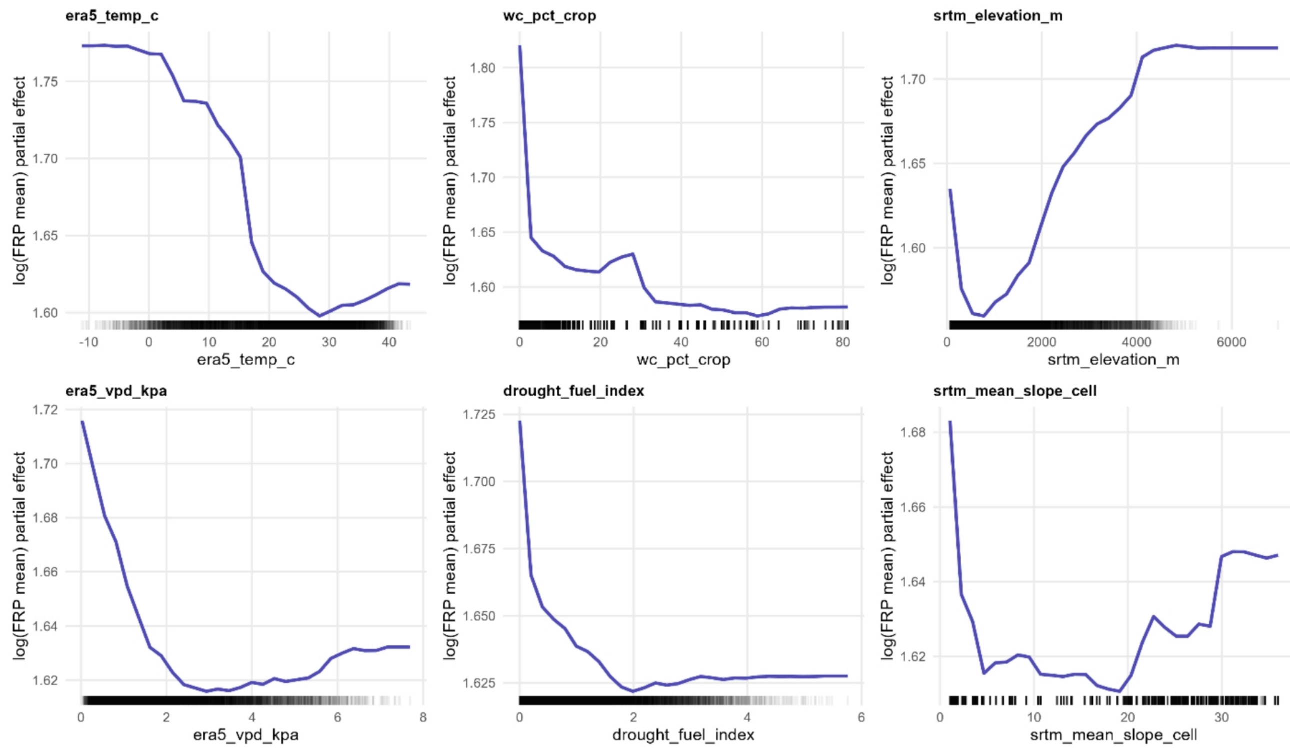

Figure 6

Random Forest partial dependence plots: Partial dependence plots for the six highest ranked predictor variables from the Random Forest regression model. Each panel shows the marginal effect of one predictor on log(FRP mean + 1) with all other predictors held at their mean values, computed using 30 equally spaced grid points across each predictor's observed range. Rug marks at the bottom of each panel indicate the observed data density. The y-axis represents the partial effect on log transformed FRP mean (MW).

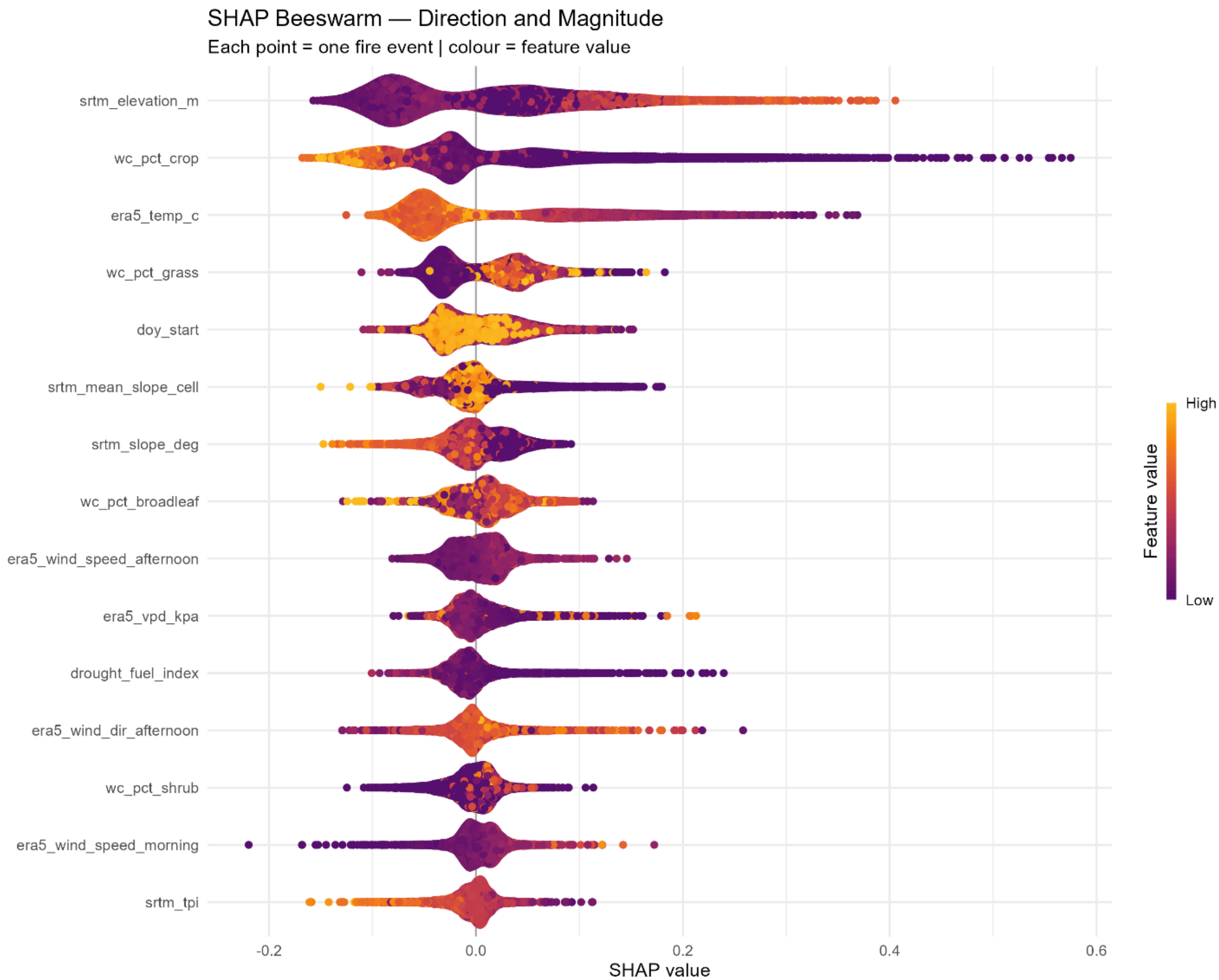

Figure 7

SHAP beeswarm plot showing the direction and magnitude of feature contributions to predicted log(FRP mean) for all 11,595 fire events. Features are ranked vertically by mean absolute SHAP value (most important at top). Each point represents one fire event; horizontal position indicates the SHAP value (positive = increases predicted FRP, negative = decreases predicted FRP); colour indicates the feature value for that event (red = high feature value, blue = low feature value). SHAP values were computed using the TreeExplainer algorithm on the final XGBoost model trained on the full dataset.

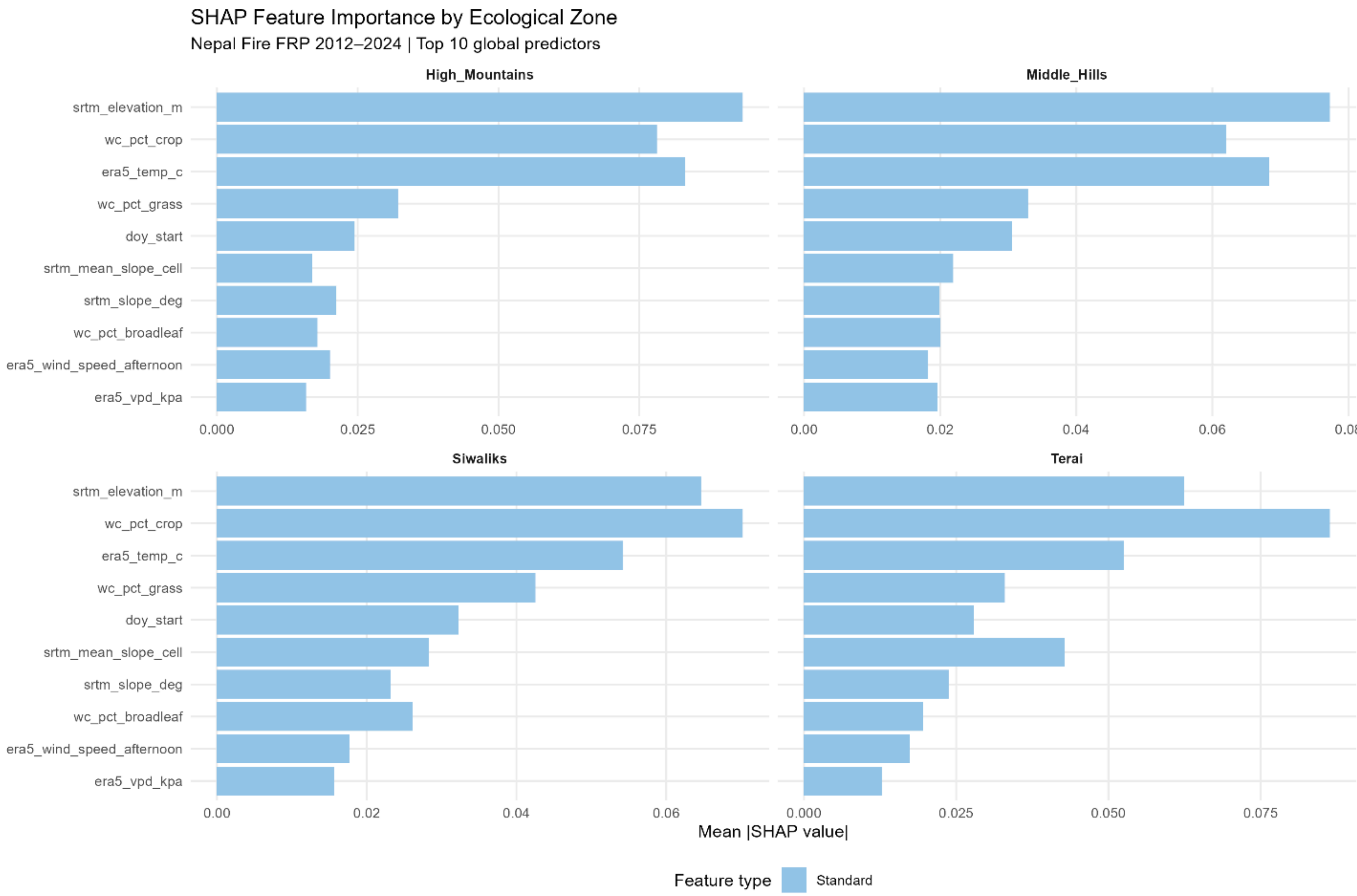

Figure 8

SHAP Feature Importance by Ecological Zone: Mean absolute SHAP values for the top 10 globally-ranked predictors, stratified by ecological zone. Each panel shows the mean |SHAP| for one zone (Terai, n = 1,218; Siwaliks, n = 3,528; Middle Hills, n = 3,696; High Mountains, n = 3,153). Purple bars indicate novel terrain–wind interaction features; blue bars indicate standard predictors. The x-axis scale varies freely between panels to show within zone ranking clearly.

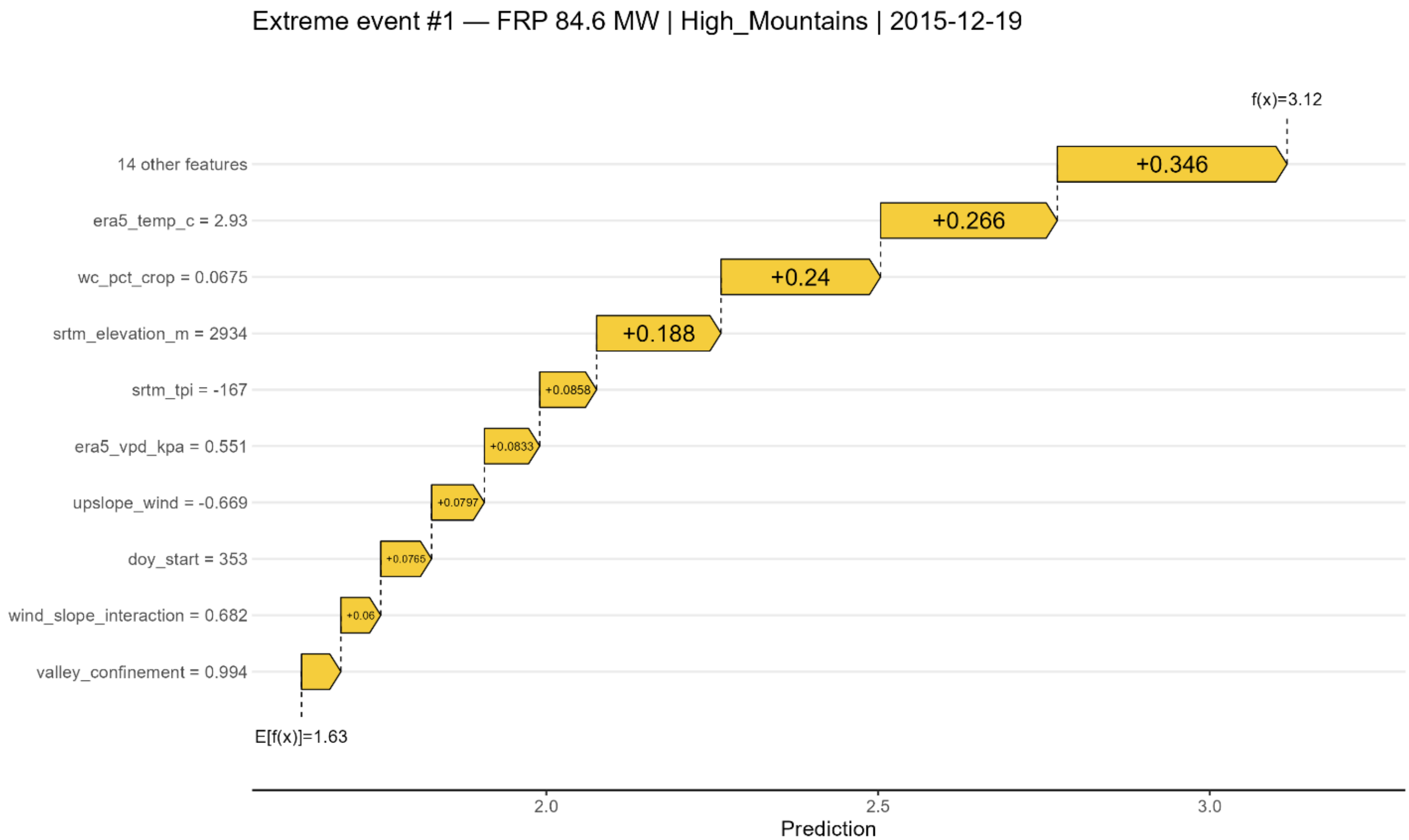

Figure 9

SHAP Extreme Waterfall Event #1: SHAP waterfall plot for the highest fire radiative power event in the Nepal 2012–2024 dataset. Each bar represents one predictor's additive contribution to the model's prediction for this event, measured as the deviation from the dataset mean prediction (E [f (x)= 1.63]). Red bars indicate features that increase predicted FRP above the mean; blue bars indicate features that decrease it. The final predicted value (f(x)=3.12) is shown at the top. This event occurred in the High Mountain zone on 2015-12-19 with a recorded peak FRP of 626.1 MW.

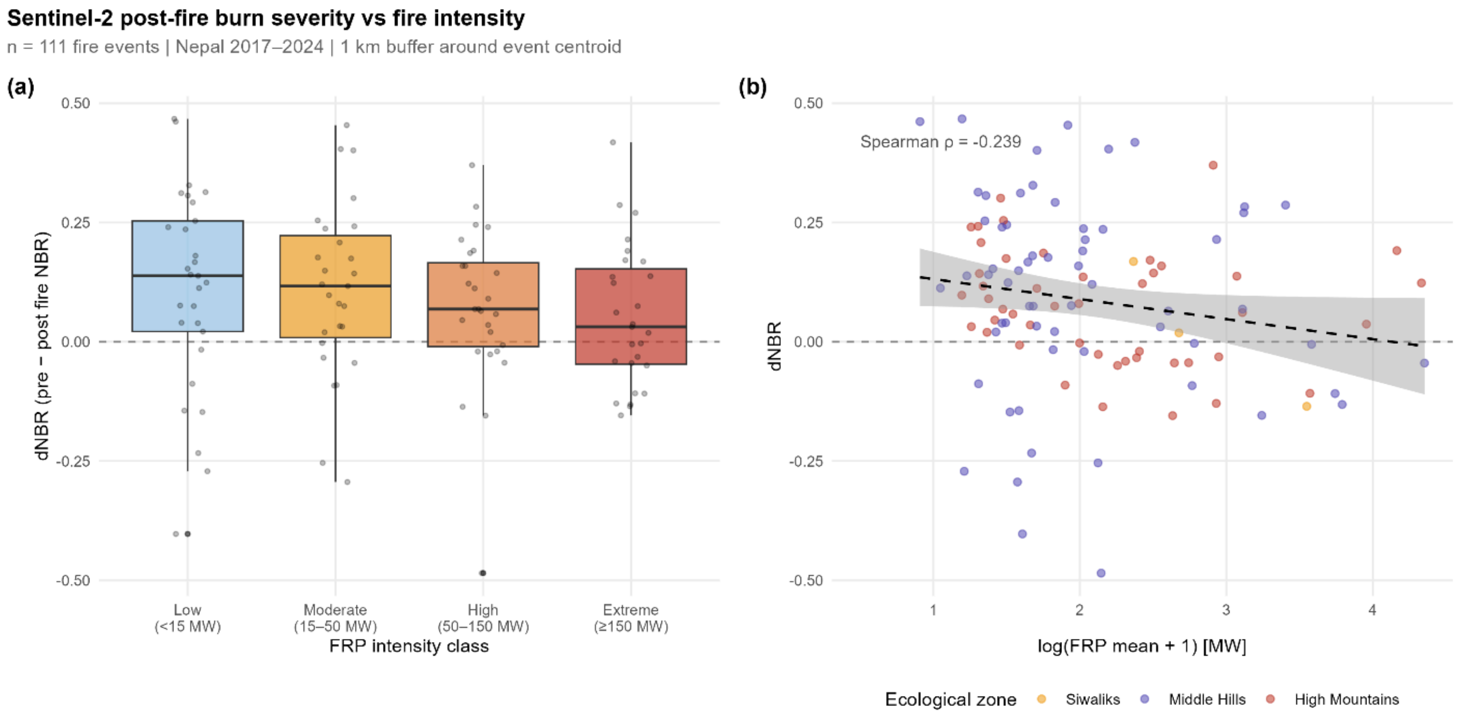

Figure 10

Sentinel-2 post fire burn severity vs fire intensity: Sentinel-2 post fire burn severity analysis for a stratified sample of Nepal fire events (n = 111, 2017–2024). (a) Boxplots of dNBR (pre fire minus post fire Normalised Burn Ratio) by FRP intensity class. The dashed horizontal line indicates dNBR = 0; positive values indicate net vegetation removal. Individual event values are overlaid as jittered points. (b) Scatter plot of log transformed FRP mean versus dNBR, coloured by ecological zone. The dashed trend line shows the ordinary least squares regression across all events (Spearman $\rho = -0.239$, $p = 0.012$). Pre-fire composites derived from Sentinel-2 SR scenes 10–60 days before fire start; post fire composites from 15–75 days after fire end; 1 km buffer around event centroid; cloud threshold <30%.